CHAPTER ONE: VOLUMETRIC QUANTITATIVE ANALYSIS

It's the experimental determination of the concentration of the substance in relation to the one with a known concentration, the latter acting as a **standard solution**. In this kind of analysis the solution of unknown concentration is pipette into a conical flask and then a suitable indicator is added, a standard solution from the burette is titrated into the conical flask until the end-point is reached. The standard solution volume is recorded in a table.

The apparatus required for volumetric analysis are; a pipette, a burette, a conical flask, two beakers, a retort stand and an indicator (a suitable indicator is used for any kind of acid-base titration). For every reaction for the neutralization on an acid, there is a suitable indicator for the reaction and below is a summary for the reaction(s) and the suitable indicators.

Acid/Base titration	**Indicators**
Strong acid and strong base	Phenolphthalein and methyl orange
Strong acid weak base	Methyl orange
Weak acid strong base	Phenolphthalein
Weak acid and weak base	Has no suitable indicator

The substances that chemists use to determine whether the substance under investigation is acidic or alkaline are called **indicators.** Different indicators show different colour changes, a summary of indicators and their colour changes in the acidic and alkaline medium (solution).

Indicator	**Medium**	
	Colour in acidic medium	Colour in alkaline medium
Phenolphthalein	Colourless	Pink
Methyl orange	Pink	Yellow
Methyl red	Red	Yellow
Blue litmus	Red	Blue
Red litmus	Red	Blue

Presentation of results.

The capacity or volume of pipette used should be written in the space provided at the top of the table of results because various pipettes exist with different capacities. The first recording is usually referred to as a trial to give one a hint of the range in which the titre values are to lie, you are required to repeat the procedure until; you obtain consistent results; the titre values should not differ by 0.20

Note:

- The eyes of the observer must be perpendicular to the level of the solution in the burette and pipette.
- The burette readings/titre values must be to two decimal places; the last decimal places being a 0 or 5. The pipette reading must be a whole number, example;

Capacity of pipette used : 25 cm^3

Titration number	1	2	3	4
Final burette reading/cm^3	25.00	25.20	38.40	27.20
Initial burette reading/cm^3	0.20	0.00	13.20	02.00
Volume of **BA2** used/cm^3	24.80	25.20	25.20	25.20

Example one

Aim : standardization of a solution of sulphuric acid using sodium carbonate.

You are provided with the following;

BA_1 a 0.05M sodium carbonate.

BA_2 sulphuric acid of unknown concentration.

You are required to determine the concentration of the acid in g/dm^3.

Procedure.

Pipette 20cm^3 or 25cm^3 of **BA_1** into a conical flask; add 2 – 3 drops of methyl orange indicator. Titrate this mixture with **BA_2** from the burette. The endpoint is when the solution just turns pink. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results:

Capacity of pipette used: 20cm^3

Titration number	1	2	3	4
Final burette reading/cm^3	13.20	26.20	39.20	13.00
Initial burette reading/cm^3	0.00	13.20	26.20	0.00
Volume of **BA_2** used/cm^3	13.20	13.00	13.00	13.00

Values of **BA_1** used to calculate the average

13.00cm^3, 13.00cm^3, 13.00cm^3

Average volume of **BA_2** used

$$\frac{13.00 + 13.00 \qquad + 13.00}{3}$$

$= 13.00cm^3$

(a) Write the equation for the reaction.

$$H_2SO_4(aq) + Na_2CO_3(aq) \longrightarrow Na_2SO_4(aq) + H_2O(i) + CO_2(g)$$

(b) Determine the number of moles;

(i) of the base.

= $1000cm^3$ of the Na_2CO_3 solution contain 0.05moles.

= $1cm^3$ of the Na_2CO_3 solution will contain $\frac{0.05}{1000}$ moles

= $20cm^3$ of the Na_2CO_3 solution will contain $\frac{0.05}{1000}$x20 moles

= 0.001moles.

(ii) Of the acid.

Form the equation of reaction in (a)

= 1 mole of H_2SO_4 (aq) neutralizes 1mole of Na_2CO_3(aq)

= therefore 0.001 x 1 = 0.001moles

(c) Determine the molarity of the acid

= $13.00cm^3$ contain 0.001moles of H_2SO_4(aq)

= $1cm^3$ will contain $\frac{0.001}{13}$moles of H_2SO_4(aq)

= $1000cm^3$ will contain $\frac{0.001}{13}$x1000moles of H_2SO_4(aq)

= 0.07M

(d) Calculate the concentration of the acid in g/dm^3.

(H = 1, S = 32, O = 16)

Relative molecular mass of H_2SO_4

= [(1 x 2) + 32 + (16 x 4)] = 98

= 0.07moles of H_2SO_4 (aq) will contain (0.07 x 98)

= concentration of the acid is 6.86g/l.

Example two

Aim: Determination of the atomic mass of M

You are provided with the following;

BA_1 which is a 0.1M sulphuric acid per litre.

BA_2 a solution of metal hydroxide **MOH** made by dissolving 8.0g per litre.

You are required to determine the atomic mass of **M** (where **M** is not a symbol of the element).

Procedure.

Pipette 20cm^3 or 25cm^3 of **BA_1** into a conical flask, add 2 – 3 drops of phenolphthalein indicator. Titrate the mixture with **BA_2** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results.

Capacity of pipette used: 20.0cm^3

Titration number	1	2	3	4
Final burette reading/cm^3	19.90	37.90	23.00	43.00
Initial burette reading/cm^3	0.00	17.90	3.00	23.00
Volume of **BA_2** used/cm^3	19.90	20.00	20.00	20.00

Values of **BA_2** used to calculate the average

20.00cm^3, 20.00cm^3, 20.00 cm^3

Average volume of **BA_2** used

$$\frac{20.00 + 20.00 + 20.00}{3} = 20.00$$

(a) Determine the number of moles of;

(i) **BA_1** that reacted.

= 1000cm^3 of **BA₁** contain 0.1moles of H_2SO_4(aq).

= 1cm^3 of **BA₁** will contain $\frac{0.1}{1000}$ moles of H_2SO_4(aq).

= 20.0cm^3 of **BA₁** solution will contain $\frac{0.1}{1000}$x20.0 moles

= 0.002moles

(ii) **BA₂** that reacted

$H_2SO_4(aq) + 2MOH(aq) \longrightarrow M_2SO_4(aq) + H_2O(i)$

From the equation 1mole of H_2SO_4(aq) reacts with 2moles of MOH

Therefore 0.002moles of H_2SO_4(aq) will react to give (0.002 x 2)moles of MOH(aq)

= 0.004moles

(b) Determine the molarity of the hydroxide

= 20cm^3 of **BA₂** will contain 0.004moles of MOH(aq)

= 1mole of **BA₂** will contain $\frac{0.004}{20}$ moles of MOH(aq)

= 1000cm^3 of **BA₂** will contain $\frac{0.004}{20.00}$ x 1000 moles of MOH(aq)

Molarity = 0.2M

(c) The atomic mass of M given that (H = 1, O = 16).

= 0.2moles of **BA₂** contain 8.0g of MOH

= 1.0mole of MOH will contain $\frac{8.0}{0.2}g$

= 40g

= M + (16 + 1) = 40

= M + 17 = 40

= M = 23g

Example three

Aim : Determination of the value of n of the acid

You are provided with the following;

BA_1 a solution containing 0.25moles of an acid **H_nA** per litre

BA_2 a solution of 0.5M sodium hydroxide

You are required to determine the basicity and the value of n of the acid.

Procedure.

Pipette 20cm^3 or 25cm^3 of **BA_2** into a conical flask, add 2 – 3 drops of methyl orange indicator. Titrate the solution with **BA_1** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results.

Capacity of pipette used: 25 cm^3

Titration number	1	2	3	4
Final burette reading/cm^3	25.80	25.20	38.40	27.20
Initial burette reading/cm^3	0.00	0.00	13.20	02.00
Volume of **BA_1** used/cm^3	25.80	25.20	25.20	25.20

Values of **BA_1** used to calculate the average

25.20cm^3, 25.20cm^3, 25.20cm^3

Average volume of **BA_1** used

$$\frac{25.20 + 25.20 + 25.20}{3}$$

= 25.20cm^3

(a) Determine the number of moles;

(i) Of sodium hydroxide

= 1000cm^3 of **BA_2** contain 0.5moles of NaOH(aq).

= 1cm^3 of **BA$_2$** will contain $\frac{0.5}{1000}$ moles of NaOH(aq).

= 25.0cm^3 of **BA$_2$** contain $\frac{0.5}{1000}$x25.0moles of NaOH(aq).

= 0.00125moles.

(ii) Of the acid.

= 1000cm^3 of **BA$_1$** contain 0.25moles of acid **H_nA**(aq)

= 1cm^3 of **BA$_1$** will contain $\frac{0.25}{1000}$ moles of acid **H_nA**(aq)

= 25.20cm^3 of **BA$_1$** will contain $\frac{0.25}{1000}$ x25.20 moles of acid **H_nA**(aq)

= 0.0063moles

b) Determine the value of n

= 0.0063moles of **H_nA** (aq) react with n moles of sodium hydroxide

= 1 mole of **H_nA** (aq) will react with $\frac{0.0125}{0.0063}$ n moles of sodium hydroxide

Therefore n = 2

Example four

Aim : Determine the percentage purity of the oxalic acid sample

You are provided with the following;

FA$_3$ a solution of 0.1M sodium hydroxide solution

FA$_4$ an aqueous solution made by dissolving 8.85gof impure oxalic acid in a litre

You are required to determine the percentage purity the oxalic acid, $H_2C_2O_4.2H_2O$

Procedure.

Pipette 20cm^3 or 25cm^3 of **FA$_3$** into a conical flask, add 2 – 3 drops of phenolphthalein indicator. Titrate the solution with **FA$_4$** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results.

Capacity of pipette used: 25 cm^3

Titration number	1	2	3	4
Final burette reading/cm^3	20.00	38.00	18.00	25.00
Initialburette reading/cm^3	0.00	20.00	0.00	7.00
Volume of **FA_4** used/cm^3	20.00	18.00	18.00	18.00

Values of **FA_4** used to calculate the average

18.00cm^3, 18.00cm^3, 18.00cm^3

Average volume of **FA_4** used.

$$\frac{18.00 + 18.00 + 18.00}{3} = 18.00cm^3$$

Write the equation for the reaction that took place.

$2NaOH(aq) + H_2C_2O_4(aq) \longrightarrow Na_2C_2O_4(aq) + 2H_2O(l)$

(a) Determine the number of moles of the;

(i) **FA_3** that reacted.

= 1000cm^3 of **FA_3** contain 0.1moles of NaOH(aq)

= 1cm^3 of **FA_3** will contain $\frac{0.1}{1000}$ moles of NaOH(aq)

= 25.0cm^3 of **FA_3** will contain moles of $\frac{0.1}{1000}$ x 25.0 NaOH(aq)

= 0.0025moles

(ii) **FA_4** that reacted.

= from the equation for the reaction that took place 2 moles of the base react with 1 mole of the acid.

1 mole of the base reacts with $\frac{1}{2}$ moles of the acid.

0.0025 moles of the base will contain $\frac{1}{2}$ x 0.0025 moles of the acid.

0.00125 moles

(b) Determine the molarity of **FA_4**.

= $18.00cm^3$ of **FA_4** contain 0.00125 moles of oxalic acid.

= $1cm^3$ of **FA_4** will contain $\frac{0.00125}{18.00}$ moles of oxalic acid.

= $1000cm^3$ of **FA_4** will contain $\frac{0.00125}{18.00}$x1000 moles of oxalic acid.

= 0.0694M

≈ 0.07M

(c) Calculate the concentration of **FA_4** in g l^{-1}

= relative molecular mass of $H_2C_2O_4.2H_2O$

= (2 x 1) + (2 x 12) + (4 x 16) + (2 x 1 x 2) + (2 x 16) = 126

= 126 x 0.07 = $8.82gl^{-1}$

(d) Calculate the percentage purity of FA4

= percentage purity = $\frac{\text{pure mass}}{\text{Impure mass}}$ x 100

= $\frac{8.82}{8.85}$ x 100

= 99.66%

Example Five

Aim: determination of the value of K, number of moles of water of crystallisation

You are provided with the following;

BA_1 an impure sample of sodium carbonate, $Na_2CO_3.\mathbf{K}H_2O$ prepared by dissolving 28.60g/dm^3

BA_2 a 0.2M monobasic acid solution

You are required to determine the value of **K**, number of moles of water of crystallisation.

Procedure.

Pipette 20cm^3 or 25cm^3 of **BA_1** into a conical flask, add 2 – 3 drops of methyl orange indicator. Titrate the solution with **BA_3** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below (Na = 23, C = 12, H =1, O = 16)

Results.

Capacity of pipette used: 25 cm^3

Titration number	1	2	3
Final burette reading/cm^3	28.00	30.00	28.50
Initial burette reading/cm^3	0.00	5.00	3.50
Volume of **BA_2** used/cm^3	28.00	25.00	25.00

Values of **BA_2** used to calculate the average

25.00cm^3, 25.00cm^3,

Average volume of **BA_2** used

$$\frac{25.00 + 25.00}{2} = 25.00cm^3$$

Write the equation for the reaction that took place

$CO_3^{2-}(aq) + 2H^+(aq) \rightarrow H_2O(l) + CO_2(g)$

(a) Determine the number of moles;

(i) **BA_1** that reacted.

= $1000cm^3$ of **BA_2** contain 0.2moles of a monobasic acid.

= $1cm^3$ of **BA2** will contain $\frac{0.02}{1000}$ moles of a monobasic acid

= $25.0cm^3$ of **BA_2** will contain $\frac{0.02}{1000}$ x 25.0 moles of a monobasic acid

= 0.005moles.

(ii) **BA_2** that reacted.

= from the equation 1 mole of **BA_1** react with 2 moles of **BA_2.**

= 1 mole of the base reacts with $\frac{1}{2}$ moles of the acid.

= 0.0025 moles of the base will contain $\frac{1}{2}$ x 0.005 moles of the acid.

= 0.0025 moles

(b) Determine the molarity of **BA_1**

= $25.00cm^3$ of **BA_1** contain 0.0025 moles of sodium carbonate.

= $1cm^3$ of **BA_1** will contain $\frac{0.0025}{25.00}$ moles of sodium carbonate.

= $1000cm^3$ of **BA_1** will contain $\frac{0.0025}{25.00}$ x 1000 moles of sodium carbonate.

= 0.1M

(c) Determine the;

(i) relative formula mass of **BA_1**.

= 0.1 moles of **BA_1** weighs 28.60g

= 1 mole will weigh $\frac{28.6}{0.1}$ g

= 286g

(ii) the value of K

= the relative molecular mass of Na_2CO_3

= (23 x 2) + 12 + (16 x 3) = 106g

= the relative molecular mass of H_2O

= (1 x 2) + 16 = 18g

= 286 – 106 = 180g

= 18g of water contain 1 mole

= 1g of water will contain $\frac{1}{18}$ moles

= 180g of water will contain $1x\frac{180}{18}$

K = 10 moles.

EXPERIMENT 1

Aim : standardization of a solution of sodium hydroxide using hydrochloric acid.

You are provided with the following;

BA_1, which is a 0.1M hydrochloric acid solution.

BA_2, which is sodium hydroxide solution of unknown concentration.

You are required to determine the concentration of the sodium hydroxide solution.

Procedure.

Pipette 20cm^3 or 25cm^3 of **BA_2** into a conical flask, add 2 – 3 drops of phenolphthalein indicator. Titrate the solution with **BA_1** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

(Na = 23, C = 12, H =1, O = 16)

Results.

Capacity of pipette used: ………………………………………………….cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of $\mathbf{BA_1}$ used/cm^3			

Values of $\mathbf{BA_1}$ used to calculate the average

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Average volume of **BA1** used

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Write the equation for the reaction that took place

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(a). Determine the number of moles;

(i) $\mathbf{BA_1}$ that reacted.

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(ii) $\mathbf{BA_2}$ that reacted.

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(b) Determine the molarity of **BA_1**

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(d) Calculate the concentration in g/dm^3.

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EXPERIMENT 2

Aim: Standardization of a solution of sulphuric acid using sodium carbonate.

You are provided with the following;

BA_1, a solution of sodium carbonate made by dissolving 10.60g in a litre.

BA_2, sulphuric acid solution of unknown concentration.

You are required to determine the concentration of the acid in g/dm^3.

Procedure.

Pipette $20cm^3$ or $25cm^3$ of **BA_1** into a conical flask, add 2 – 3 drops of methyl orange indicator. Titrate the solution with **BA_2** from the burette. The end point is when the solution just turns pink. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results.

Capacity of pipette used: ……………………………………………….….cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of **BA_2** used/cm^3			

Values of **BA_2** used to calculate the average

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Average volume of **BA_2** used

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a) Write the ionic equation for the reaction that took place

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(b) Determine the molarity of the sodium carbonate that reacted.

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(c) Determine the number of moles of;

(i) The sodium carbonate

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(ii) sulphuric acid.

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(d) Determine the molarity of the acid.

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(e) Calculate the concentration of the acid in g/dm^3.

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Eperiment 3

You are provided with the following;

BA_1 which is a 0.1M sulphuric acid per litre.

BA_2 a solution of metal hydroxide $Q(OH)_2$ made by dissolving 4.64g per litre.

You are required to determine the atomic mass of **Q** (where **Q** is not a real symbol of the element).

Pipette $20cm^3$ or $25cm^3$ of **BA_2** into a conical flask, add 2 – 3 drops of methyl orange indicator. Titrate the mixture with **BA_1** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

TABLE OF RESULTS:

Volume of pipette used……………………..cm^3

Titration number	1	2	3
Final burette reading (cm^3)			
Initial burette reading (cm^3)			
Volume of $\mathbf{BA_1}$ used (cm^3)			

Volumes of $\mathbf{BA_1}$ used to calculate the average volume.

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Average volume of $\mathbf{BA_1}$ used.

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QUESTIONS:

a) Determine the number of moles of

(i) $\mathbf{BA_2}$ that reacted.

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(ii) $\mathbf{BA_1}$ that was reacted.

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b) Determine the molarity of the hydroxide.

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c) The atomic mass of Q given that (H = 1, O = 16).

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EXPERIMENT 4

You are provided with the following:

BA_1 which is a 0.1M sodium hydroxide solution

BA_2, which is a solution made by dissolving 12.5g of $XSO_4.5H_2O$ in $1dm^3$

BA3, which is a 0.2M hydrochloric acid.

You are required to determine the value of n in $XSO_4.5H_2O$.

a) Pipette 25cm3 or 20cm3 of BA1 into a clean conical flask
b) Using a measuring cylinder, transfer 25cm3 of BA2 into a conical flaskin (a) Shake the contents for about one minute and allow to stand. Label the resultant solution BA4.
c) Titrate the solution BA4 using BA3 from a burette while shaking gently until a precipitate dissolves to form a pale green solution.
d) Repeat the procedures (a) to (c) until you get consistent results Record your results in the table below.

TABLE OF RESULTS:

Volume of pipette used……………………………………………………………….cm^3

Titration numbers	1	2	3
Final burette reading (cm^3)			
Initial burette reading (cm^3)			
Volume of **BA_3** used (cm^3)			

Volumes of **BA_3** used to calculate the average volume.

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Average volume of **BA_3** used.

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QUESTIONS:

(a) Determine the number of moles of

i. Hydrochloric acid in **BA_3**. (H=1, O=16, Na=23).

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ii. $X(OH)_2$ in BA4 that reacted with BA3. (1 mole of $X(OH)_2$ reacts with 2 moles of hydrochloric acid)

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iii. Moles of $XSO_4.5H_2O$ that reacted.

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iv. $XSO_4.5H_2O$ in a litre of solution.

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b) Determine the,

i. Mass of one mole of $XSO_4.5H_2O$

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ii. The value of X in $XSO_4.5H_2O$

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Experiment 5.

Aim : Standardization of a solution of sulphuric acid using sodium carbonate.

You are provided with the following;

BA_1, which is a 0.05M sodium carbonate

BA_2 , which is sulphuric acid solution of unknown concentration.

You are required to determine the concentration of the acid in g/dm^3.

Procedure.

Pipette 20cm^3 or 25cm^3 of **BA_1** into a conical flask, add 2 – 3 drops of methyl orange indicator. Titrate the solution with **BA_2** from the burette. The end point is when the solution just turns pink. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results.

Capacity of pipette used : …………………………………………………cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of **BA_2** used/cm^3			

Values of **BA_2** used to calculate the average

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Average volume of **BA_2** used

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a) Write the ionic equation for the reaction that took place

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(b) Determine the number of moles of;

(i) the **BA_1.**

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(ii) Of the **BA_2.**

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(d) Determine the molarity of the acid.

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(e) Calculate the concentration of the acid in g/dm^3.

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EXPERIMENT 6

Aim: standardization of sodium carbonate using a solution of sulphuric acid.

You are provided with the following;

BA1, which is sodium carbonate of unknown concentration

BA_1, which is 0.1M sulphuric acid solution.

You are required to determine the concentration of the alkali in g/l.

Procedure.

Pipette $20cm^3$ or $25cm^3$ of **BA_2** into a conical flask, add 2 – 3 drops of methyl orange indicator. Titrate the solution with **BA_1** from the burette. The end point is when the solution just turns pink. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results.

Capacity of pipette used: ……………………………………………………cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of **BA_1** used/cm^3			

Values of **BA_1** used to calculate the average

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Average volume of **BA_1** used

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a) Write the ionic equation for the reaction that took place

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b) Determine the number of moles of;

(i) the acid.

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(ii) the base.

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c) Determine the molarity of the sodium carbonate that reacted.

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(d) Determine the molarity of the sodium carbonate.

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(e) Calculate the concentration of the sodium carbonate in g/dm^3.

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EXPERIMENT 7

Aim: determination of the atomic mass of Z

You are provided with the following;

BA_1, which is a solution made by dissolving 3.45g of a metal carbonate, Z_2CO_3 in 250 cm^3 of distilled water.

BA_2, which is a 0.05M solution of sulphuric acid per litre.

You are required to determine the atomic mass of element Z (Z is not a symbol of the element).

Procedure.

Pipette 20 cm^3 or 25 cm^3 of **BA_2** into a conical flask, add 2 – 3 drops of methyl orange indicator. Titrate the mixture with **BA_1** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results.

Capacity of pipette used : ………………………………………………….cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of **BA_1** used/cm^3			

Values of **BA_1** used to calculate the average

……………………………………………………………………………………cm^3

Average volume of **BA_1** used

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a) Determine the number of moles of

(i) **BA_2** that reacted

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ii) **BA_1** that reacted

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b) Determine the molarity of **BA_1**

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....

....

....

....

(c) Determine the formula mass of the carbonate, Z_2CO_3

....

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....

(d) Determine the the relative atomic mass of Z

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EXPERIMENT 8

Aim: Determination of the atomic mass of M

You are provided with the following

BA_1, which is a 0.1M sulphuric acid per litre

BA_2, which is a solution of a metal hydroxide MOH made by dissolving 8.00g per litre

You are required to determine atomic mass of M (where M is not the real symbol of the element).

Procedure

Pipette 20cm^3 or 25cm^3 of **BA_1** into a conical flask, add 2 – 3 drops of methyl orange indicator. Titrate the mixture with **BA_2** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results

Capacity of pipette used: ....cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of **BA_2** used/cm^3			

Values of **BA_2** used to calculate the average titre

....

Average volume of **BA_2** used

....

....

(a) Determine the number of

(i) **BA_1** that reacted

....

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....

(ii) **BA_2** that reacted

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(b) Molarity of the hydroxide

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....

(c) Determine the atomic mass of M given that H = 1, O = 16.

....

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EXPERIMENT 9

Aim: determination of the value of n of the acid

You are provided with the following;

BA_1, which is a solution containing 2.80g potassium hydroxide per $500cm^3$.

BA_2, which is a solution containing 9.00g of the acid HnX per litre.

You are required to determine the value of n

Procedure.

Pipette $20cm^3$ or $25cm^3$ of **BA_1** into a conical flask, add 2 - 3 drops of phenolphthalein indicator.

Titrate the mixture with **BA_2** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results.

Capacity of pipette used: ………………………………………………….cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of **BA_2** used/cm^3			

Values of **BA_2** used to calculate the average

……………………………………………………………………………………… cm^3

Average volume of **BA_2** used

………

…… cm^3

Write the ionic equation for the reaction

………

………

(a) Determine the number of moles of;

(i) **BA_2**that reacted.

………

………

………

(ii) **BA_2** that reacted (RFM of H_nX = 90).

………

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(b) Determine the reaction ratio.

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(c) State the value of n

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………

(d) What is the basicity of H_nX

………

EXPERIMENT 10

Aim : Determination of the value of n of the acid

You are provided with the following;

BA_1 is a solution containing 0.25 moles of an acid H_nA per litre.

BA_2 is a solution of 0.5M sodium hydroxide.

You are required to determine the basicity and value of n of the acid

Procedure.

Pipette 20 cm^3 or 25 cm^3 of **BA_2** into a conical flask, add 2 – 3 drops of phenolphthalein indicator.

Titrate the mixture with **BA_1** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results.

Capacity of pipette used : …………………………………………………cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of **BA_1** used/cm^3			

Values of BA1 used to calculate the average

………………………………………………………………………………………….cm^3

Average volume of **BA_1** used

……

…… cm^3

(a). Determine the number of moles of the;

(i) sodium hydroxide.

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(ii) Acid

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(b) Determine the value of n.

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EXPERIMENT 11

Aim: Determination of the value of n of the acid

You are provided with the following;

BA_1, which is a 0.2M sodium hydroxide solution.

BA_2 which is a solution made by dissolving 12.60g of $H_2Y.nH_2O$

You are required to determine the value of n

Procedure.

Pipette 20cm^3 or 25cm^3 of **BA_2** into a conical flask, add 2 – 3 drops of phenolphthalein indicator.

Titrate the mixture with **BA_1** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results.

Capacity of pipette used : ……………………………………………………cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of **BA_1** used/cm^3			

Values of **BA_1** used to calculate the average

……………………………………………………………………………………….cm^3

Average volume of **BA_1** used

………

……… cm^3

(a)Determine the number of moles of the;

(i) Sodium hydroxide that reacted.

………

………

………

………

(ii) Acid that reacted.

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………

(b) Determine the molarity of **BA_2**.

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(c) Calculate the

(i) Molar mass of the acid.

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(ii) The value of n given that (Y = 88, O = 16, H = 1)

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EXPERIMENT 12

Aim: determination of the basicity of acid K

You are provided with the following;

BA_1, which is a 0.05M solution of an unknown acid.

BA_1, which is a 0.1M sodium hydroxide solution.

You are required to determine the basicity of acid K.

Procedure.

Pipette 20cm3 or 25cm3 of **BA_1** into a conical flask, add 2 – 3 drops of phenolphthalein indicator. Titrate the mixture with **BA_2** from the burette. The endpoint is when the solution just turns colourless. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results.

Capacity of pipette used: ………………………………………………….cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of **BA_1** used/cm_3			

Values of **BA_1** used to calculate the average

....cm^3

Average volume of **BA_1** used

....

....

(a) Determine the number of moles of;

(i) **BA_1** that reacted.

....

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(ii) **BA_2** that reacted.

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(b) Determine the basicity of the acid.

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EXPERIMENT 13

Aim: determination of the relative formula mass of compound XCO_3

You are provided with the following;

BA_1, a solution made by dissolving 2.50g of a metal carbonate XCO_3 in 250cm^3 of distilled water.

BA_2, a 0.1M hydrochloric acid solution.

You are required to determine the relative formula mass of compound XCO_3 and the atomic mass of X.

Procedure.

Pipette 20 cm^3 or 25 cm^3 of **BA_1** into a conical flask, add 2 – 3 drops of phenolphthalein indicator. Titrate the mixture with **BA_2** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

Results.

Capacity of pipette used : ……………………………………………….cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of **BA_1** used/cm^3			

Values of **BA_1** used to calculate the average

……………………………………………………………………………….cm^3

Average volume of **BA_1** used

………

…………………………………………………………………………………………………… cm^3

(a) Determine the number of moles of the;

(i) Hydrochloric acid that reacted.

……………………………………………………………………………………………

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……………………………………………………………………………………………

(ii) the metal carbonate XCO_3 that reacted.

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……………………………………………………………………………………………………

(b) Determine the relative formula mass of the carbonate XCO_3.

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……………………………………………………………………………………………………

(c) Determine the relative atomic mass of X

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EXPERIMENT 14

Aim : Determination of the percentage purity of a sodium carbonate sample

You are provided with the following;

BA_1, which is a 0.1M solution of sulphuric acid.

BA_2, which is a solution of an impure sample of anhydrous sodium carbonate made by dissolving 23.40g per litre.

You are required to determine the percentage purity of the sodium carbonate sample.

Procedure.

Pipette $20cm^3$ or $25cm^3$ of **BA_1** into a conical flask, add 2 – 3 drops of methyl orange indicator. Titrate the mixture with **BA_2** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

(Na = 23, C = 12, O = 16, S = 32, H = 1)

Results.

Capacity of pipette used: ……………………………………………cm^3

Titration number	1	2	3

Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of $\mathbf{BA_1}$ used/cm^3			

Values of $\mathbf{BA_1}$ used to calculate the average

……………………………………………………………………………………cm^3

Average volume of $\mathbf{BA_1}$ used

………

…… cm^3

Write the equation for the reaction that took place.

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………

(a) Determine the number of moles of;

(i) $\mathbf{BA_1}$ that reacted.

………

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(ii) $\mathbf{BA_2}$ that reacted.

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………

(b) Determine the molarity of $\mathbf{BA_2}$

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……….

(c) Calculate the;

(i) Concentration of the sodium carbonate in g/dm^3

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(ii) Percentage purity of the sample.

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EXPERIMENT 15

Aim : determination of the percentage purity of a potassium hydroxide sample

You are provided with the following;

BA_1, a solution containing 54.5g of potassium hydroxide and potassium sulphate

BA_2, a 0.5M solution of sulphuric acid.

You are required to determine the percentage of the potassium hydroxide in **BA_1**.

Procedure.

(i) Measure and record the temperature of **BA_1**

(ii) Pipette 20cm^3 or 25cm^3 of **BA_1** into a plastic beaker

(iii)Measure 10cm3 of **BA_2** using a measuring cylinder and transfer it at once into a plastic beaker containing **BA_1**. Gently stir with a thermometer and record the highest temperature attained by the mixture.

(iv)Repeat procedure (ii) and (iii) using 15,20, 25, 30 and 35cm3 of **BA_2**.

(v) . Record your results in the table below.

Results.

Capacity of pipette used: ....cm^3

Volume of **BA_2** used(cm^3)	0	10	15	20	25	30	35
Temperature of solution(^0C)							

Questions

(a) Plot a graph of solution mixture against volume of **BA_2** used.

(b) Determine from your graph, the maximum volume of **BA_2** required to react with **BA_1**

....

....

(c) Calculate the

(i) Maximum number of moles of sulphuric acid that reacted.

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....

(ii) The number of moles of potassium hydroxide in 20.0cm3 (25.0cm3) of **BA_1**

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....

(iii) concentration of **BA_1** in grams of potassium hydroxide per litre of solution. K = 39, O = 16, , H = 1)

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(iv) percentage of potassium sulphate in **BA_1**.

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EXPERIMENT 16

Aim : Determination of the percentage purity of the oxalic acid sample

You are provided with the following;

BA_3, a 0.1M solution of sodium hydroxide.

BA_4, an aqueous solution of oxalic acid made by dissolving 8.50g in a litre.

You are required to determine the percentage purity of the oxalic acid sample.

Procedure.

Pipette $20cm^3$ or $25cm^3$ of **BA_3** into a conical flask, add 2 – 3 drops of phenolphthalein indicator. Titrate the mixture with **BA_4** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below. (C = 12, O = 16, S = 32, H = 1)

Results.

Capacity of pipette used: …………………………………………………cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of **BA_4** used/cm^3			

Values of **BA_4** used to calculate the average

………………………………………………………………………………………cm^3

Average volume of **BA_4** used

………

………………………………………………………………………………………………cm^3

Write the equation for the reaction that took place.

………

………

(a) Determine the number of moles of;

(i) Sodium hydroxide that reacted.

………

………

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………

(ii) The oxalic acid that reacted.

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………

(b) Determine the molarity of **BA_4**

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………

(c) Calculate the;

(i) concentration of the **BA_4** in g/l

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....

(ii). percentage purity of the **BA_4**.

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EXPERIMENT 17

You are provided with the following.

BA_1, a solution prepared by dissolving 7.15g of sodium carbonate, $Na_2CO_3.YH_2O$ in 250 cm^3 of distilled water.

BA_2, a **0.2M** hydrochloric acid solution

You are required to determine the value of Y, number of moles of water of crystallization in the hydrated compound, $Na_2CO_3.YH_2O$.

Pipette $20cm^3$ or $25cm^3$ of **BA_1** into a conical flask, add 2 – 3 drops of methyl orange indicator. Titrate the mixture with **BA_2** from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below

TABLE OF RESULTS

Capacity of pipette used: ....cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			

Volume of **BA_2** used/cm^3			

Values of **BA_2** used to calculate the average

....cm^3

Average volume of **BA_2** used

....cm^3

Write the equation for the reaction that took place.

....

....

a) Determine the number of moles of;

(i) **BA_2** that reacted.

....

....

....

(ii) **BA_1** that reacted.

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b) Determine the

(i) concentration of Na_2CO_3 in moles/dm^3

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(ii) Molar mass of $Na_2CO_3.YH_2O$.

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……………………………………………………………………………………………………

……………………………………………………………………………………………………

(iii)Value of Y

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Experiment 18

You are provided with the following;

BA_1 which is a 0.2 hydrochloric acid per litre.

BA_2 a solution of metal hydroxide **$M(OH)_2$** made by dissolving 7.4 g per litre.

You are required to determine the atomic mass of **M** (where **M** is not a symbol of the element).

Pipette 20cm^3 or 25cm^3 of BA_1 into a conical flask, add 2 – 3 drops of methyl orange indicator. Titrate the mixture with BA_2 from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below.

TABLE OF RESULTS:

Volume of pipette used…………………….cm^3

Titration number	1	2	3
Final burette reading (cm^3)			
Initial burette reading (cm^3)			
Volume of **BA_1** used (cm^3)			

Volumes of **BA_1** used to calculate the average volume.

……………………………………………………………………………………cm^3

BA_1 used.

Average volume of

....

....cm^3

QUESTIONS:

a) Determine the number of moles of

(i) **BA_1** that reacted.

....

....

....

(ii) **BA_2** that was reacted.

....

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....

b) Determine the molarity of **BA_2.**

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c) The molar mass of **$M(OH)_2$**

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d) The atomic mass of **M** given that (H = 1, O = 16).

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Experiment 20

You are provided with the following

BA1 which is 0.05M sodium carbonate solution.

BA2 which is a $0.1MH_nB$.

You are required to determine the basicity of the acid.

Pipette $20cm^3$ or $25cm^3$ of $\mathbf{BA_1}$ into a conical flask, add 2 – 3 drops of methyl orange indicator. Titrate the mixture with $\mathbf{BA_2}$ from the burette. Repeat the procedure to obtain consistent results. Record your results in the table below

TABLE OF RESULTS

Capacity of pipette used: ....cm^3

Titration number	1	2	3
Final burette reading/cm^3			
Initial burette reading/cm^3			
Volume of $\mathbf{BA_2}$ used/cm^3			

Values of $\mathbf{BA_2}$ used to calculate the average

....cm^3

Average volume of $\mathbf{BA_2}$ used

....cm^3

Write the equation for the reaction that took place.

....

....

c) Determine the number of moles of;

(iii)**Acid, H_nB** that reacted.

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(iv)**Sodium carbonate** that reacted.

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………

d) Determine the

(i) The mole ratio of the acid and sodium carbonate

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(i) The value of n in H_nB

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Chapter 2: Qualitative analysis

Introduction.

These notes are just a summary of the key reactions which can easily be carried out in a test tube to give you a clue on what is contained in the sample.

A sheet of paper printed, that gives a clue about the sample. The sheet is divided into three columns column with three sub headings i.e.

1. Test **2.Observation** **3.deduction**

1. Test

This column consists of all the tests to be carried out on the unknown substance, it's fully written up for you to read, understand and follow the instruction.

2. Observation.

Under this column you record all your observations (what you have seen) i.e. colour changes (from the original colour of the unknown) and the colour change that occurred when performing the experiment in their order of appearance, precipitates or solutions, gas(es) evolved and their smell.

You should completely identify the gas being evolved record the tests carried out on a particular gas in the observation column.

You might have missed an observation in a particular test, at the end of the test after getting an idea of the unknown check your observations and deductions to fill what you had not observed.

3. Deduction.

From your theory you can deduce about the identity of the unknown, in this column.

Qualitative analysis deals with the identification of the chemical composition of compounds in terms of ions that is

Cations (positively charged ions) they are;

Al^{3+}, Pb^{2+}, Zn^{2+}, Mg^{2+}, Ca^{2+}, NH_4^+, Fe^{2+}, Cu^{2+}, Fe^{3+}

Anions (negatively charged ions)

CO_3^{2-}, Cl^-, SO_4^{2-}, NO_3^-, CO_3^{2-}

Here we are concerned with only the tests used for the identification of simple ions many of which you have come across in your theory, so you must be familiar with the chemical reactions involving the ions handled at ordinary level and identifying them.

Tests on common substances

By taking correct observations on performing the test below one can correctly identify the ions (cations and anions) present in a given sample.

1. Appearance of the sample

Here our main focus lies on the colour and nature of the substance.

Observation	Deduction
White powder/white crystals	Al^{3+}, Pb^{2+}, Zn^{2+}, Mg^{2+}, Ca^{2+}, NH_4^+
Colourless crystals/colourless solution	Al^{3+}, Pb^{2+}, Zn^{2+}, Mg^{2+}, Ca^{2+}, NH_4^+
Blue powder/blue crystals/blue solution	Fe^{2+}, Cu^{2+}
Yellow/brown crystals/yellow/brown solution	Fe^{3+}
Green powder/green crystals/green solution	Fe^{2+}, Cu^{2+}

2. Action on heating

The unknown sample is heated in the boiling tube (hard glass test tube) until there is no further observable change and the gases given off are tested. The sample must be placed in a dry test tube; if the test tube is not dry then it's inclined at an angle of 45^o to the burner with the rim/mouth facing down.

- In almost each case a gas is evolved. It must be tested with a named reagent of your choice to confirm the gas being evolved.

- No mark is awarded for any deductions unless the observation(s) is /are **correct** and **complete;** i.e. each correct observation must have a correct corresponding deduction.
- In the identification of ions, a mark is awarded only when the symbols of ions are correctly written and confirmed at the right stages.

Tests of common gases

When a gas is given off, it must be tested with a named reagent of your choice to confirm it. The table below gives a clue about the gas and there possible tests.

Observation	Deduction
(i) i) a colourless liquid which turns anhydrous copper sulphate blue (i) A colourless liquid seen on cooler parts of the test tube.	Water of crystallization
(ii) A colourless gas turns lime water milky and litmus red	CO_2(g) from CO_3^{2-} CO_2(g) from CO_3^{2-}
(iii)i) A colourless gas turns litmus red and decolourises (turns form purple to colourless) purple potassium permanganate ($KMnO_4$)	SO_2(g) from SO_4^{2-}
ii) white fumes of a gas turn litmus red	SO_3(g) from SO_4^{2-}
iii) A colourless gas with a chocking, irritating smell turns litmus red and acidified potassium dichromate ($K_2Cr_2O_7$) from orange to green.	SO_2(g) from SO_4^{2-}
(iv)Reddish-brown fumes turn litmus red	NO_2(g) from NO_3^-
(v) Chocking irritating colourless gas turns litmus blue, forms white fumes with concentrated hydrochloric acid on a glass rod.	NH_3(g) from NH_4^+

(vi)a yellowish green gas, turns litmus red and bleaches it.	$Cl_2(g)$ from Cl^-
(vii) colourless gas turns blue litmus red and decolourises acidified purple potassium permanganate ($KMnO_4$)	H_2S gas
(viii) Chocking smell of a colourless gas turns litmus red and acidified potassium dichromate ($K_2Cr_2O_7$) from orange to green.	H_2S

The table below shows what happens on further heating of the unknown sample leading to the formation of oxides

Observation	**Deduction**
White sublimate.	NH_4^+
Residue turned from yellow to white on cooling.	Zn^{2+} ions present, ZnO remains
Yellow residue remained yellow when cold.	Pb^{2+} ions present, PbO remains
Reddish brown	Fe_2O_3
Brown residue	Fe_2O_3
Black residue	FeO, CuO

- If the substance given contains two cations the residue is more likely to give two colours each corresponding to a particular cation, but if there is only one cation the residue will have only one colour on cooling.
- For cases were many colours are seen focus must mainly go to the darkest colour(s).
- In some cases red litmus may turn blue then red on strong heating of the unknown substance. This means that there are two colourless gases being given off one alkaline and the other acidic.

3. **Dissolving the sample**

- Where the sample partly dissolves forming a filtrate or a residue; the filtrate contains Cl^-, SO_4^{2-} and NO_3^-,
- When dilute nitric acid or dilute hydrochloric acid or concentrated sulphuric acid is added to the residue "effervescence of a colourless gas, turns lime water milky and litmus red."CO_2(g) from CO_3^{2-}, CO_3^{2-} confirmed.
- If the sample readily dissolves in water the ions present are Cl^-, SO_4^{2-}, NO_3^-,

More information on the solubility of salts

- All nitrates are soluble.
- All chlorides are soluble except AgCl and $PbCl_2$
- All sulphates are soluble except $BaSO_4$ and $PbSO_4$
- All carbonates are insoluble except Na_2CO_3, K_2CO_3 and $(NH_4)_2CO_3$
- All hydroxides are insoluble except KOH, NaOH and NH_4OH

4. **Reaction with sodium hydroxide solution**

The sodium hydroxide solution is used in the identification of cations in the unknown substance, sodium hydroxide is added to the unknown solution drop wise and later added until in excess with frequent shaking and then the observation is immediately taken/ noted on paper. Note;

- The colour of the precipitate or solution if any is formed when the sodium hydroxide is added drop wise and later in excess.
- Whether the precipitate formed persists or dissolves in the excess of the reagent.

Observation	**Deduction**
White precipitate, soluble in excess	Al^{3+}, Zn^{2+}, Pb^{2+}, $Al^{3+}(aq) + 3OH^-(aq) \rightarrow Al(OH)_3(s)$

	$Zn^{2+}(aq) + 2OH^{-}(aq) \rightarrow Zn(OH)_2(s)$ $Pb^{2+}(aq) + 2OH^{-}(aq) \rightarrow Pb(OH)_2(s)$
White precipitate, insoluble in excess	Ca^{2+}, Mg^{2+}, $Ca^{2+}(aq) + 2OH^{-}(aq) \rightarrow Ca(OH)_2(s)$ $Mg^{2+}(aq) + 2OH^{-}(aq) \rightarrow Mg(OH)_2(s)$
No observable change, a colourless gas with a chocking smell turns red litmus blue, forms white fumes with concentrated hydrochloric acid on a glass rod.	NH_4^{+} confirmed $NH_4^{+}(aq) + OH^{-}(aq) \rightleftharpoons NH_4OH(aq)$ On warming $NH_4OH(aq) \rightarrow NH_3(g) + H_2O(l)$
Blue precipitate Insoluble in excess	Cu^{2+} $Cu^{2+}(aq) + 2OH^{-}(aq) \rightarrow Cu(OH)_2(s)$
Dirty green precipitate, insoluble in excess	Fe^{2+}, $Fe^{2+}(aq) + 2OH^{-}(aq) \rightarrow Fe(OH)_2(s)$
Reddish-brown precipitate, insoluble in excess	Fe^{3+} $Fe^{3+}(aq) + 2OH^{-}(aq) \rightarrow Fe(OH)_3(s)$

- The hydroxides of aluminium, zinc and lead dissolve in excess sodium hydroxide solution to form complex ions because they are amphoteric hydroxides.

Equations for the complex ions formed

$Al(OH)_3(s) + OH^{-}(aq) \rightarrow [Al(OH)_4]^{-}(aq)$

$Pb(OH)_2(s) + 2OH^{-}(aq) \rightarrow [Pb(OH)_4]^{2-}(aq)$

$Zn(OH)_2(s) + 2OH^{-}(aq) \rightarrow [Zn(OH)_4]^{2-}(aq)$

- After tests with sodium hydroxide solution for the test where a white precipitate, insoluble in excess indicating the presence Ca^{2+} and Mg^{2+} are suspected.

5. **Reaction with ammonium hydroxide solution.**

Ammonium hydroxide solution is common known as, ammonia solution / aqueous ammonia because the ionization of ammonia in water is a reversible process; the ammonium hydroxide formed is unstable so it forms ammonia and water.

Equation $NH_3(g) + H_2O(l) \rightleftharpoons NH_4OH(aq$

Observation	Deduction
White precipitate, insoluble in excess	Al^{3+}, Pb^{2+} $Pb^{2+}(aq) + 2OH^-(aq) \rightarrow Pb(OH)_2(s)$ $Al^{3+}(aq) + 3OH^-(aq) \rightarrow Al(OH)_3(s)$
White precipitate, soluble in excess to form a colourless solution	Zn^{2+} confirmed Equations below*
White precipitate, insoluble in excess(ppt previously insoluble in NaOH)	Mg^{2+} $Mg^{2+}(aq) + 2OH^-(aq) \rightarrow Mg(OH)_2(s)$
No observable change	Ca^{2+} $Ca^{2+}(aq) + 2OH^-(aq) \rightarrow Ca(OH)_2(aq)$
Dirty green precipitate, insoluble in excess	Fe^{2+}(its confirmed if it's the last test on the Fe^{2+} sample) $Fe^{2+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_2(s)$
Reddish-brown precipitate, insoluble in excess	Fe^{3+} (its confirmed if there is no other test on the Fe^{3+} sample) $Fe^{3+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_3(s)$
Blue precipitate dissolves in excess forming a deep blue solution.	Cu^{2+} confirmed Equations below*

- The hydroxides of zinc and copper dissolve in excess ammonia solution to form complex ions.

Equations.*

$Zn^{2+}(aq) + 2OH^-(aq) \rightarrow Zn(OH)_2(s)$

Zinc hydroxide.

$Zn(OH)_2(s) + 4NH_3(aq) \rightarrow [Zn(NH_3)_4]^{2+}(aq) + OH^-(aq)$

Tetra ammine zinc (II) ions

$Cu^{2+}(aq) + 2OH^{-}(aq) \rightarrow Cu(OH)_2(s)$

Copper hydroxide

$Cu(OH)_2(s) + 4NH_3(aq) \rightarrow [Cu(NH_3)_4]^{2+}(aq) + OH^{-}(aq)$

Tetra ammine copper (II) ions.

- The reactions of the given unknown samples with sodium hydroxide are not the complete identification of any particular ion.
- Fe^{2+} and Fe^{3+}, Cu^{2+} and Zn^{2+} ions are confirmed using ammonia solution.

Confirmatory tests

Identification of cations

Lead (II) Pb^{2+}

To the unknown solution that initially had a white precipitate insoluble in excess ammonia solution

- Add potassium iodide solution, forms a yellow precipitate.
  $Pb^{2+}(aq) + 2I\text{-}(aq) \rightarrow PbI_2(s)$
- Add dilute hydrochloric acid, a white precipitate, soluble on heating and in concentrated hydrochloric acid
  $Pb^{2+}(aq) + 2Cl(aq) \rightarrow PbCl_2(s)$
- Add dilute sulphuric acid; a white precipitate, insoluble in excess.
  $Pb^{2+}(aq) + SO_4^{2-}(aq) \rightarrow PbSO_4(aq)$

Aluminium, Al^{3+}

To the unknown solution that initially formed a white precipitate insoluble in excess ammonia solution.

- Add potassium iodide solution, the solution remains clear.
  $Al^{3+}(aq)+3I(aq) \rightarrow AlI_3(aq)$

Iron (II), Fe^{2+}

To the unknown solution;

- Add ammonia solution drop wise until in excess, dirty green precipitate, insoluble in excess and later turns brown on standing, (it's confirmed if it's the last test on the Fe^{2+} sample)

  $Fe^{2+}(aq) + 2OH^{-}(aq) \rightarrow Fe(OH)_2(s)$
- Add a few drops of potassium hexacyanoferatte (II) solution; a pale blue precipitate is formed.
- Add a few drops of potassium hexacyanoferatte(III) solution; a dark blue precipitate is formed.
- Add a few drops of ammonium or potassium thiocyanate solution; there is no observable change.

Iron (III), Fe^{3+}

To the unknown solution;

- Add ammonia solution drop wise until in excess; brown precipitate, insoluble in excess, (It's confirmed if it's the last test on the Fe^{3+} sample)

  $Fe^{3+}(aq) + 3OH^{-}(aq) \rightarrow Fe(OH)_3(s)$
- Add a few drops of potassium hexacyanoferatte (II) solution; a dark blue precipitate.
- Add a few drops of potassium hexacyanoferatte (III) solution; there is no observable change.
- Add a few drops of ammonium potassium thiocyanate solution; a deep red solution.

Zinc (II) Zn^{2+}

To the unknown solution;

- Add ammonia solution drop wise until in excess; white precipitate, soluble in excess

  $Zn^{2+}(aq) + 2OH^{-}(aq) \rightarrow Zn(OH)_2(s)$

  $Zn(OH)_2(s) + 4NH_3(aq) \rightarrow [Zn(NH_3)_4]^{2+}(aq) + OH^{-}(aq)$

  Tetra ammine zinc (II) ions

Ammonium, NH_4^+

To the unknown solution;

- Add sodium hydroxide solution drop wise until in excess and warm the mixture; no observable change, a colourless gas with a chocking smell turns red litmus blue, forms white fumes with concentrated hydrochloric acid on a glass rod.

$NH_4^+(aq) + OH\text{-}(aq) \rightleftharpoons NH_4OH(aq)$

On warming **$NH_4OH(aq) \rightarrow NH_3(g) + H_2O(l)$**

Copper (II) Cu^{2+}

To the unknown solution;

- Add ammonia solution drop wise until in excess; blue precipitate dissolves in excess forming a deep blue solution

  $Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$

  $Cu(OH)_2(s) + 4NH_3(aq) \rightarrow [Cu(NH_3)_4]^{2+}(aq) + 2OH\text{-}(aq)$

  Tetra ammine copper (II) ions

- Add a few iron filings to a solution, blue solution turns pale green and brown solid deposited

  $Cu^{2+}(aq) + Fe(s) \longrightarrow Fe^{2+}(aq) + Cu(s)$

Summary for the Identification of cations

Test	**Observation**	**Deduction**
To the unknown solution that initially had a white precipitate insoluble in excess ammonia solution. Add potassium iodide solution.	Yellow precipitate	Pb^{2+} confirmed
	Solution remain clear (no observable change)	Al^{3+} confirmed
To the unknown solution; add a few drops of potassium hexacyanoferate(II) solution.	A deep blue precipitate	Fe^{3+} confirmed
	A pale blue precipitate	Fe^{2+} confirmed
To the unknown solution, Add a	A dark blue precipitate.	Fe^{2+} confirmed

few drops of potassium hexacyanoferate(III) solution,	There is no observable change (colour of the solution remains)	Fe^{3+} confirmed
To the unknown solution, Add a few drops of ammonium potassium thiocyanate solution,	There is no observable change.	Fe^{2+} confirmed
	Deep red solution	Fe^{3+} confirmed
To the unknown solution; add ammonia solution drop wise until in excess	White precipitate, insoluble in excess	Zn^{2+} confirmed
To the unknown solution; add sodium hydroxide solution drop wise until in excess and warm the mixture.	No observable change, a colourless gas with a chocking smell turns red litmus blue, forms white fumes with concentrated hydrochloric acid on a glass rod.	NH_3(g) from NH_4^+ NH_4^+ confirmed
To the unknown solution; add ammonia solution drop wise until in excess	Blue precipitate dissolves in excess forming a deep blue solution	Cu^{2+} confirmed
• Add a few iron filings to a solution,	blue solution turns pale green and brown solid deposited	Cu^{2+} ions reduced to copper, and iron oxidized to Fe^{2+}

Identification of anions

1. **The use of lead(II)nitrate solution.**

It's very vital in the detection of the presence or absence of the insoluble lead salts mainly CO_3^{2-}, Cl^- and SO_4^{2-}. If lead nitrate is added to the unknown aqueous solution, the following is observed and deduced;

- If there is a white precipitate formed then CO_3^{2-}, Cl^-, SO_4^{2-} are suspected present.
- On warming if the precipitate formed dissolves and reappears on cooling, Cl^- confirmed.
- If the precipitate formed persists on the addition of an acid the Cl^- and SO_4^{2-} are suspected.
- If the precipitate formed dissolves on addition of dilute nitric acid, Effervescence of a colourless gas, turns lime water milky and litmus red, CO_3^{2-} confirmed.

2. **The use of either barium chloride or barium nitrate solution(s)**

It's also for the detection of the presence of insoluble barium salts mainly CO_3^{2-} and SO_4^{2-}. If a barium nitrate or barium chloride solution is added to the unknown aqueous solution the following is observed and deduced;

- If there is a white precipitate formed then CO_3^{2-} and SO_4^{2-} are suspected present.
- On addition of an acid, if the white precipitate is insoluble, then a SO_4^{2-} is confirmed.

3. **Addition of silver nitrate solution.**

Add silver nitrate solution to the unknown aqueous solution, a white precipitate is observed, Cl^- confirmed.

4. **Use of freshly prepared Iron (II) sulphate followed by conc. sulphuric acid**

To the unknown solution add freshly prepared iron (II) sulphate, slant a test tube and carefully add concentrated sulphuric acid **don't shake,** a brown ring is observed, NO_3^-, confirmed.

Summary for the Identification of anion

Test	Observation	Deduction
(a) (i) Add lead (II) nitrate solution.	White precipitate	CO_3^{2-}, SO_4^{2-}, Cl^-, $Pb^{2+}(aq) + CO_3^{2-}(aq) \rightarrow PbCO_3(s)$ $Pb^{2+}(aq) + SO_4^{2-}(aq) \rightarrow PbSO_4(s)$ $Pb^{2+}(aq) + Cl^-(aq) \rightarrow PbCl_2(s)$
ii) On warming	White precipitate, Insoluble	CO_3^{2-}, SO_4^{2-}, $Pb^{2+}(aq) + CO_3^{2-}(aq) \rightarrow PbCO_3(s)$ $Pb^{2+}(aq) + SO_4^{2-}(aq) \rightarrow PbSO_4(s)$
	White precipitate dissolves and reappears on cooling.	Cl^- confirmed $Pb^{2+}(aq) + 2Cl^-(aq) \rightarrow PbCl_2(s)$
	White precipitate, Insoluble	SO_4^{2-}, Cl^-, $Pb^{2+}(aq) + SO_4^{2-}(aq) \rightarrow PbSO_4(s)$ $Pb^{2+}(aq) + Cl^-(aq) \rightarrow PbCl_2(s)$
iii) If followed by nitric acid	Effervescence of a colourless gas, turns lime water milky and litmus red,	CO_3^{2-} confirmed
(b) add barium nitrate solution	White precipitate	CO_3^{2-}, SO_4^{2-}, $Ba^{2+}(aq) + CO_3^{2-}(aq) \rightarrow BaCO_3(s)$ $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$
If followed by an acid	White precipitate, Insoluble in acid.	SO_4^{2-} confirmed $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$
(c) add silver nitrate solution.	White precipitate	Cl^- confirmed $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
(d) Add freshly prepared iron (II) sulphate, slant a test tube and carefully add	A brown ring observed.	NO_3^-, confirmed.

concentrated sulphuric acid.		

To distinguish between the precipitate, lead chloride, lead carbonate, and lead sulphate after the addition of lead (II) nitrate.

- Lead chloride is soluble on warming and reappears on cooling.
- Lead carbonate is soluble in acid with effervescence of a colourless gas that turns lime water milky.
- Lead sulphate is insoluble in acid.

Nitrate ions.

- There instances we have a white precipitate when ferrous sulphate is added to a solution suspected to have nitrate ions this is mainly because of the lead ions combining with the sulphate ions to form insoluble lead sulphate which is the white precipitate in the excess ferrous sulphate. A student should be very observant because the brown ring is very hard to observe depending on the concentration of the iron (II) sulphate.
- In the other instance there is no observable change but on adding the concentrated sulphuric acids the brown ring is formed.

Simple tests on the preliminary experiments

Reaction of sodium hydroxide solution with different cations in solution and the equations for the reaction(s) that take place.

Ion(cation)	Observation(s) and equation(s)
Ca^{2+}, Mg^{2+},	
Cu^{2+}	

Al^{3+}, Zn^{2+}, Pb^{2+},	
Fe^{2+},	
Fe^{3+}	
NH_4^+	

Reaction of ammonium hydroxide (aqueous ammonia) solution with different cations in solution and the equations for the reaction(s) that take place.

Ion(cation)	Observation(s) and equation(s)
Ca^{2+},	

Mg^{2+},	
Al^{3+}, Pb^{2+},	
Fe^{2+},	
Fe^{3+}	
Cu^{2+}	
NH_4^+	

Example 1

Test	Observation	Deduction
(a) Dissolve a spatula end full of **Q** in $5cm^3$ of distilled water. Shake vigorously, filter. Keep both the filtrate and the residue. Divide the resulting solution into five portions.	White powder/crystalline partially dissolves, pale green filtrate and white residue.	Al^{3+}, Pb^{2+}, Zn^{2+}, Mg^{2+}, Ca^{2+}, NH_4^+ Cu^{2+}, Fe^{2+}, Cl^-, SO_4^{2-}, NO_3^-, in the filtrate CO_3^{2-} residue

(b) To the first portion of the filtrate, add sodium hydroxide solution drop wise until in excess and warm the mixture.	Dirty green precipitate, insoluble in excess, a colourless gas with a chocking smell turns red litmus blue, forms white fumes with concentrated hydrochloric acid on a glass rod.	Fe^{2+} NH_3(g) from NH_4^+ NH_4^+ confirmed
ii) To the second portion add ammonia solution drop wise until in excess.	Dirty green precipitate, insoluble in excess,	Fe^{2+} confirmed
iii) To the third portion add a few drops of dilute nitric acid followed by lead(II) nitrate	White precipitate,	Cl^-, SO_4^{2-},
iv) To the fourth portion add barium nitrate solution	White precipitate,	SO_4^{2-} present
v) To the fifth portion add a few drops of nitric acid followed by silver nitrate solution.	Solution remains clear	SO_4^{2-} confirmed
(c) Heat a spatula end full of the residue in a hard test tube strongly until there's no further change.	a colourless gas turns lime water milky and litmus red, Residue turned yellow when hot and white on cooling.	CO_2(g) from CO_3^{2-} ZnO
(d) Divide the remaining part of the residue in a	Effervescence of a colourless gas turns lime	CO_2(g) from CO_3^{2-} CO_3^{2-} confirmed

minimum amount of dilute nitric acid, Divide the resulting solution into two portions.	water milky and litmus red, colourless solution.	
(i) To the first portion of the filtrate, add sodium hydroxide solution drop wise until in excess.	White precipitate, soluble in excess forming a colourless solution	Al^{3+}, Pb^{2+}, Zn^{2+}, probably present.
ii) To the second portion add ammonia solution drop wise until in excess.	White precipitate, soluble in excess forming a colourless solution	Zn^{2+} confirmed

Identify the; cations; NH_4^+, Fe^{2+}, Zn^{2+}

Anion; SO_4^{2-}, CO_3^{2-}

Example 2

You are provided with substance **D12** which contains two cations and an anion. Carryout the following tests on the solid **D12** to identify the cations and anion in **D12**. Identify the gases evolved; record your observations and deductions in the table below

Test	Observation	Deduction
(a) Heat a spatula end full of **D12** strongly in a hard test tube until there is no further change.	Blue crystalline solid, a colourless gas with a chocking smell turns red litmus blue, forms white fumes with concentrated hydrochloric acid on a glass rod, colourless gas turns litmus and	Cu^{2+}, Fe^{2+}, NH_3(g) from NH_4^+ SO_2(g) from SO_4^{2-} Water of crystallization, NH_4^+ salt, CuO FeO probably formed as residue

	decolourises acidified potassium permanganate ($KMnO_4$) a colourless liquid seen on cooler parts of the test tube, white sublime and a black residue.	
(b) Dissolve a spatula end full of **D12** in $3cm^3$ of distilled water. Divide the resulting solution into three portions	Fully dissolves to form a blue solution	Cu^{2+}, Fe^{2+}, Cl^-, SO_4^{2-}, NO_3^-,
i) To the first portion add sodium hydroxide solution drop wise until in excess and warm the mixture.	Blue precipitate, insoluble in excess, a colourless gas with a chocking smell turns red litmus blue, forms white fumes with concentrated hydrochloric acid on a glass rod.	Cu^{2+} NH_3(g) from NH_4^+ NH_4^+ confirmed
ii) To the second portion add ammonia solution drop wise until in excess.	Blue precipitate dissolves in excess forming a deep blue solution.	Cu^{2+} confirmed
iii) To the third portion add lead (II) nitrate solution and warm the mixture.	White precipitate, insoluble in excess	SO_4^{2-}, CO_3^{2-},
iv) To the fourth portion add barium nitrate	White precipitate, insoluble in acid	SO_4^{2-}, confirmed

solution followed by an acid.		

Identify the; cations ; Cu^{2+}, NH_4^+ and Anion ; SO_4^{2-},

Example 3

You are provided with substance **R** which contains one cation and an anion. Carryout the following tests on the solid **R** to identify the cation and anion in **R**. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat a spatula end full of **R** strongly in a hard test tube until there is no further change. (note the appearance)	Green powder gives out a colourless gas turns lime water milky and litmus red, black solid left	Cu^{2+}, CO_2(g) from CO_3^{2-} CuO is the black solid
(b) Dissolve a spatula end full of **R** in concentrated sulphuric acid. Divide the resulting solution into three portions	Effervescence of a colourless gas, turns lime water milky and litmus red, blue solution.	CO_2(g) from CO_3^{2-} CO_3^{2-} confirmed Cu^{2+} present
i) To the first portion add sodium hydroxide solution drop wise until in excess.	Blue precipitate, insoluble in excess	Cu^{2+} present
ii) To the second portion add ammonia solution	Blue precipitate dissolves in excess forming a deep	Cu^{2+} confirmed

drop wise until in excess.	blue solution	

Identify the; cation; Cu^{2+} Anion; CO_3^{2-}

Write the equation for the reaction that took place in b (ii).

$Cu^{2+}(aq) + 2OH^{-}(aq) \rightarrow Cu(OH)_2(s)$

Copper hydroxide

$Cu(OH)_2(s) + 4NH_3(aq) \rightarrow [Cu(NH_3)_4]^{2+}(aq) + OH-(aq)$

Tetra ammine copper (II) ions

Example 4

You are provided with substance **K2** which contains one cation and an anion. Carryout the following tests on the solid **K2** to identify the cation and anion in **K2**. Identify the gases evolved; record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat a spatula end full of **K2** strongly in a hard test tube until there is no further change. (note the appearance)	White powder, a colourless gas turns lime water milky and litmus red, orange residue when hot and yellow when cold.	Al^{3+}, Pb^{2+}, Zn^{2+}, Mg^{2+}, Ca^{2+}, NH_4^+ $CO_2(g)$ from CO_3^{2-} PbO
(b) Dissolve a spatula end full of **K2** in dilute nitric acid. Divide the resulting solution into three portions	Effervescence of a colourless gas, turns lime water milky and litmus red, clear solution.	$CO_2(g)$ from CO_3^{2-} CO_3^{2-} confirmed
i) To the first portion add	White precipitate,	Al^{3+}, Pb^{2+}, Zn^{2+},

sodium hydroxide solution drop wise until in excess.	soluble in excess	
ii). To the second portion add ammonia solution drop wise until in excess.	White precipitate, Insoluble in excess	Al^{3+}, Pb^{2+},
iii). To the third portion carryout a test of your choice to confirm the cation present in **K2**.	On addition of potassium iodide, a yellow precipitate observed.	Pb^{2+} confirmed

Identify the; cation; Pb^{2+}

Anion; CO_3^{2-}

Write the equation for the reaction that took place in b (ii).

$Pb^{2+}(aq) + 2OH^-(aq) \rightarrow Pb(OH)_2(s)$

Experiment 1

You are provided with substance D which contains one cation and an anion. Carryout the following tests on the solid D to identify the cation and anion in D. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat strongly a spatula endful of D in a hard test tube until there is no further change.		
(b) Dissolve a spatula end full of D in 3 cm^3 of distilled water. Divide the		

resulting solution into three portions		
i) To the first portion add sodium hydroxide solution drop wise until in excess.		
ii) To the second portion add ammonia solution drop wise until in excess.		
iii) To the third portion add barium nitrate solution followed by nitric acid.		

Identify the; cation ……………………………………………………………………

Anion ……………………………………………………………………

Write the equation for the reaction that took place in b (i).

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Experiment 2

You are provided with substance **D** which contains one cation and two anions. Carryout the following tests on the solid **D** to identify the cations and anions in **D**. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
a) Heat a spatula endful of **D** strongly in a hard test tube until there is no further change.		

(b) Dissolve a spatula end full of **D** in 5cm^3 of distilled water. Divide the resultant solution into four portions.		
i) To the first portion of add sodium hydroxide solution drop wise until in excess.		
ii) To the second portion add lead (II) nitrate solution and warm the mixture and leave to cool.		
iii) To the third portion add barium nitrate solution followed by dilute nitric acid. Filter and divide filtrate into 2 portions **a** and **b**		
iv) To portion **a**, add lead (II)nitrate solution followed by nitric acid.		
(v) To portion **b** add silver nitrate solution followed by ammonia solution		
(vi) To the fourth portion add dilute nitric acid followed by 3 drops of silver nitrate. Filter and keep the filtrate as portion **c.**		
(vii) To portion **c** add an equal		

volume of dilute nitric acid. Then add 3 drops of barium nitrate		

c) Identify the

(i) Cation...

(ii) Anions..

d) Explain your observation in test b (iii)

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Experiment 3

You are provided with substance **O** which contains a common cation and an anion. Carryout the following tests on the solid **O** to identify the cation and anion in **O**. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat a spatula end full of **O** strongly in a hard test tube until there is no further change.		
(b) Dissolve two spatula end full of **O** in $5cm^3$ of distilled water. Divide the resulting solution into six portions		
(ix) To the first portion of add sodium hydroxide solution		

drop wise until in excess.		
ii) To the second portion add ammonia solution drop wise until in excess.		
iii) To the third portion, add sodium carbonate followed by nitric acid.		
(iv)To the fourth portion add an equal volume of freshly prepared iron (II) sulphate, slant a test tube and carefully add $2cm^3$ concentrated sulphuric acid.		
(v) To the fifth portion, add 2 drops of dilute hydrochloric acid and warm the test tube.		
(vi)To the sixth portion, add 2 drops of potassium iodide solution		

c) Identify the

(i) Cation………………………………………………………………………………………

(ii) Anion………………………………………………………………………………………

d) Explain your observation in test b(v)

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Experiment 4

You are provided with substance **C** which contains two cations and an anion. Carryout the following tests on the solid **C** to identify the cations and anion in **C**. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat a spatula end full of C strongly in a hard test tube until there is no further change.		
(b) Put a spatula end full of **C** in a test tube. Add sodium hydroxide solution and heat gently.		
(c) Put a spatula endful of **C** in a test tube. Add $2cm^3$ of dilute nitric acid and warm, add $5cm^3$ of distilled water. Divide the resulting solution into five portions.		
(i) To the first portion of add sodium hydroxide solution drop wise until in excess and heat gently		
(ii) To the second portion add		

ammonia solution drop wise until in excess.		
(iii) To the third portion add a few drops of potassium iodide solution.		
(iv) To the fourth portion add lead(II) nitrate solution.		
(v) To the last portion add three drops of dilute barium nitrate solution followed by nitric acid.		

Identify the; cations………………………………………………………………………

Anion…………………………………………………………………………

Experiment 5

You are provided with substance **B** which contains a common cation and two anions. Carryout the following tests on the solid **B** to identify the cation and anion in **B**. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat a spatula end full of **B** in a dry test tube gently ad then strongly until there is no further change.		
(b) Dissolve a spatula end full of **B** in $5cm^3$ of distilled water. Filter and keep both filtrate		

and residue. Divide the Filtrate into four portions		
(i) To the first portion of the filtrate add sodium hydroxide solution drop wise until in excess.		
(ii) To the second portion add ammonia solution drop wise until in excess.		
(iii) To the third portion of filtrate, add a few zinc granules and leave to stand for 5 minutes.		
(iv) To the fourth portion add an equal volume of freshly prepared iron (II) sulphate, slant a test tube and carefully add 1cm^3 concentrated sulphuric acid.		
c) (i) Put some residue in a test tube. Heat gently and then strongly till there is no change		
(ii) Dissolve the		

remaining residue in about $3cm^3$ dilute nitric acid		

d) Identify the

(i) cation...

(ii) anion...

e) (i) Write the equation for the reaction that took place in b (iii).

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(ii) Explain your observation in test b (iii)

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Experiment 6

You are provided with substance **W** which contains one cation and one anion. Carryout the following tests on the solid **W** to identify the cations and anion in **W**. Identify the gases evolved, record your observations and deductions in the table below.

Test	**Observation**	**Deduction**
(a) Heat a spatula end full of W strongly in a hard test tube until there is no further change.		

b) To a spatula endful of **W** in a test tube, add $1cm^3$ of concentrated sulphuric acid and warm.		
c) Dissolve a spatula end full of **W** in $5cm^3$ of distilled water and divide the resultant solution into five portions.		
(i) To the first portion add sodium hydroxide solution drop wise until in excess.		
(ii) To the second portion add ammonia solution drop wise until in excess.		
(iii) To the third portion add a few drops of potassium hexacyno ferrate(III) solution.		
(iv) To the fourth portion, add lead (II) nitrate solution and heat the mixture and allow to cool.		
(v) To the fifth portion, add a few drops silver nitrate solution, followed by nitric acid.		

d) Identify the

(i) cation……………………………………………………………………………………………

(ii) Anion…………………………………………………………………………………………

(e) Write the equation for the reaction that took place in c (ii).

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Experiment 7

You are provided with substance E which contains one cation and one anion. Carryout the following tests on the solid E to identify the cation and anion in E. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat a spatula end full of E strongly in a hard test tube until there is no further change.		
(b) Dissolve a spatula end full of E in 5cm^3 of distilled water. Divide the filtrate into four portions.		
(i) To the first portion of add sodium hydroxide solution drop wise until in excess and heat.		

(ii) To the second portion add ammonia solution drop wise until in excess.		
(iii)To the third portion add lead (II) nitrate solution and warm the mixture.		
(iv)To the fourth portion carryout a test of your choice to confirm the anion present in E.		

Identify the; cations……………………………………………………………………

Anion.……………………………………………………………………

Write the equation for the reaction that took place in b (ii).

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Experiment 8

You are provided with substance K which contains a common cation and an anion. Carryout the following tests on the solid K to identify the cation and anion in K. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat a spatula end full of K strongly in a		

hard test tube until there is no further change.		
(b) Dissolve a spatula end full of K in 3cm^3 of distilled water. Divide the resulting solution into three portions		
(i) To the first portion of add sodium hydroxide solution drop wise until in excess.		
(ii) To the second portion add ammonia solution drop wise until in excess.		
(iii) To the third portion add freshly prepared iron (II) sulphate, slant a test tube and carefully add concentrated sulphuric acid.		

Identify the; cations…………………………………………………………………

Anions…………………………………………………………………

Write the equation for the reaction that took place in b (ii).

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Experiment 9

You are provided with a substance **M** which contains two cations and one anion. You are required to identify the ions in **M** by carrying out the following tests;

Record your observations and deductions in the table below;

TESTS	OBSERVATIONS	DEDUCTIONS
a) Strongly heat a spatula endful of **M** in a dry clean test tube until there is no further change.		
b) To a spatula end full of **M** add about $5cm^3$ of water. Shake the mixture strongly and filter. Keep both the filtrate and residue.		
c) To the residue from (b) add dilute nitric acid drop by drop until no further change (warm if necessary). Divide the resultant solution into two parts.		
(i) To the first part, add sodium hydroxide solution dropwise until in excess.		
(ii) To the second part, add ammonia solution drop wise until in excess		

d) Divide the filtrate into five parts. (i) To the first part, add sodium hydroxide solution dropwise until in excess.		
(ii) To the second part, add ammonia solution dropwise until in excess.		
(iii) Use the third part to carry out a test of your own choice to confirm one of the cations in **M**		
(iii) To the fourth part add lead (II) nitrate solution. Warm the mixture.		
(iv) Use the fifth part to carry out a test of your own choice to confirm the anion in M.		

e) Identify the

(i) Cation:…………………………………………………………… and

(ii) Anion:…………………………………………………………….

Experiment 10

You are provided with substance V which contains two cations and one anion. You are required to identify the ions in V by carrying out the tests below. Identify any gases evolved. Record your observations and deductions in the table below;

TESTS	OBSERVATIONS	DEDUCTIONS
a) To two spatula endfulls of V add about 4 cm^3 of dilute nitric acid. Warm the mixture.		
b) To the resultant solution from (a) add ammonia solution dropwise until in excess. Filter the mixture. Keep both the filtrate and residue.		
c) To the filtrate add dilute nitric acid drop by drop until the solution is just acidic. Divide the acidic solution into two parts. (i) To the first part add sodium hydroxide solution drop by drop until in excess.		
(ii) To the second part add ammonia solution drop by drop until in excess.		
a) To the residue from (b) add dilute nitric acid until no further change. Divide the resultant solution		

into three parts.		
(i) To the first part add sodium hydroxide solution drop by drop until in excess.		
(ii) To the second part add ammonia solution drop by drop until in excess.		
(iii) Use the third part to carry out a test of your own choice to confirm one of the cations in V.		

b) Identify the

(i) Cations ……………………………..and …………………………….

(ii) Anion ……………………………………………………………..

Experiment 11

You are provided with substance R which contains a common cation and an anion. Carryout the following tests on the solid R to identify the cation and anion in R. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat a spatula end full of R strongly in a hard test tube until there is no further change.(note the appearance of R)		
(b) Dissolve a spatula end full of R in 3cm^3 of dilute hydrochloric. Divide the resulting solution into three portions		
i) To the first portion of add sodium hydroxide solution drop wise until in excess.		
ii) To the second portion add ammonia solution drop wise until in excess.		
iii) To the third portion carryout a test of your choice to confirm the cation present in R		

Identify the; cation……………………………………………………………………

Anion……………………………………………………………………

Experiment 12

You are provided with substance P which contains two cations and an anion. Carryout the following tests on the solid P to identify the cations and anion in P. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a). heat a spatula end full of P strongly in a hard test tube until there is no further change.		
(b) dissolve a spatula end full of P in 5cm3 of distilled water. Divide the resulting solution into four portions		
i) To the first portion of add sodium hydroxide solution drop wise until in excess and warm.		
ii) To the second portion add ammonia solution drop wise until in excess.		
iii) To the third portion add lead (II) nitrate solution and warm the mixture.		

iv) To the fourth portion add three drops of dilute hydrochloric acid followed by barium chloride solution.		

Identify the; cations………………………………Anion.……………………………………

Experiment 13

You are provided with substance T which contains two cations and two anions. Carryout the following tests on the solid T to identify the cations and anions in T. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat a spatula end full of T strongly in a hard test tube until there is no further change.		
(b) Dissolve a spatula end full of T in $5cm^3$ of distilled water and shake vigorously, filter keep both the filtrate and the residue. Divide the resulting solution into four portions		

i) To the first portion add sodium hydroxide solution drop wise until in excess.		
ii) To the second portion add ammonia solution drop wise until in excess.		
iii) To the third portion add a few drops of nitric acid followed by barium nitrate.		
iv) To the fourth portion add lead (II) nitrate solution and warm the mixture.		
(d) Wash the residue with distilled water, add dilute nitric acid and divide the resulting solution into three portions.		
i) To the first portion add sodium hydroxide solution drop wise until in excess.		
ii) To the second portion add ammonia solution drop wise until in excess.		
iii) To the third portion		

add potassium iodide solution.		

Identify the; cations…………………………………… Anions…………………………………

Write the equation for the reaction that took place in c (ii

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Experiment 14

You are provided with substance W which contains one cation and one anion. Carryout the following tests on the solid W to identify the cations and anion in W. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat a spatula end full of W strongly in a hard test tube until there is no further change.		
b) To a spatula endful of W in a test tube, add $1cm^3$ of concentrated sulphuric acid and warm.		
c) Dissolve a spatula end full of W in $5cm^3$ of distilled water and divide the resultant solution into five portions.		

i. To the first portion add sodium hydroxide solution drop wise until in excess.		
ii. To the second portion add ammonia solution drop wise until in excess.		
ii. To the third portion add a few drops of potassium hexacyno ferrate(III)solution.		
v. To the fourth portion, add lead (II) nitrate solution and heat the mixture and allow to cool.		
v. To the fifth portion, add a few drops silver nitrate solution, followed by nitric acid.		

Identify the; cation…………………………………………………………………………

Anions…………………………………………………………………………

Write the equation for the reaction that took place in c (ii).

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Experiment 15

You are provided with substance J which contains two cations and one anion. You are required to carry out the following tests on J to identify the cations and anion in J. Identify the gas(es) evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat two spatula endfuls of J strongly in a hard test tube and allow the residue to cool. Keep the residue		
(b) Dissolve the residue in (a) in dilute hydrochloric acid. Add dilute sodium hydroxide solution drop wise until there is no further change the filter. Keep both filtrate and residue.		
(c) To the filtrate add dilute hydrochloric acid dropwise until the the solution is just acidic. Divide the solution into two parts		
i) To the first portion of add sodium hydroxide		

solution dropwise till excess.		
ii) To the second portion add ammonia solution drop wise until in excess.		
e) Dissolve the rest of the residue in dilute hydrochloric acid and divide the solution into two portions.		
i) To the second portion of the solution add dilute ammonia solution drop-wise until excess.		

Identify the; cations……………………………………………………………………

Anion ……………………………………………………………………

Experiment 16

You are provided with substance Y which contains a common cation and two anions. Carryout the following tests on the solid Y to identify the cation and anions in Y. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat a spatula end full of Y strongly in a hard test tube until		

there is no further change.		
(b) Dissolve a spatula end full of Y in $5cm^3$ of distilled water. And shake vigorously, filter keep both the filtrate and the residue. Divide the resulting solution into four portions		
(c) Wash the residue formed in (b) then dissolve it in dilute hydrochloric acid and divide the resulting solution into two portions.		
i) To the first portion of add sodium hydroxide solution drop wise until in excess.		
ii) To the second portion add ammonia solution drop wise until in excess.		
(d) Divide the filtrate into three portions. (i) To the first portion add barium nitrate		

solution.		
(ii) To the second portion of the filtrate add lead (II) nitrate solution and warm the mixture.		
(iii) To the third portion of the filtrate add a few drops of dilute silver nitrate followed by aqueous ammonia solution.		

Identify the; cations………………………………………………………………………

Anion ………………………………………………………………………………………..

Experiment 17

You are provided with substance K which contains two cations and one anion. Carryout the following tests on the solid K to identify the cations and anion in K. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
a) heat a spatula end full of K strongly in a hard test tube until there is no further change.		

b) Dissolve 3 spatula endfuls of E in $5cm^3$ of dilute nitric acid. Divide the resulting solution into four portions.		
i) To the first portion of the solution, add dilute sodium hydroxide solution drop wise until in excess.		
ii) To the second portion add ammonia solution drop wise until in excess.		
iii). To the third portion add potassium iodide solution		
c) To the fourth portion of the solution to add about 2cm3 of sodium sulphate solution and filter. Divide the filtrate into two parts.		
i) To the first parts, add dilute sodium hydroxide solution drop wise until in excess.		
ii) To the second part of		

the solution, add dilute ammonia solution drop wise until in excess.		

Identify the; cations…………………………………………………………………………

Anion ……………………………………………………………………………

Experiment 18

You are provided with substance E which contains two cations and one anion. Carryout the following tests on the solid E to identify the cations and anion in E. Identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
a) Heat a spatula end full of E strongly in a hard test tube until there is no further change.		
b) dissolve a spatula end full of E in 4cm^3 of distilled water. Divide the resulting solution into four portions.		
i) To the first portion of add sodium hydroxide solution drop wise until in excess.		

ii) To the second portion add ammonia solution drop wise until in excess.		
iii) To the third portion add lead (II) nitrate solution and warm the mixture.		
iv) To the fourth portion add three drops of dilute HCl followed by barium chloride solution.		

Identify the; cations…………………………………………………………………………

Anion ……………………………………………………………………………………………

Experiment 19

You are provided with substance **W** which contains two cations and two anions. Carryout the following tests to identify the cations and anions in **W**. Identify any gases that may be evolved. Record your observations and deductions in the table below.

TESTS	OBSERVATION	DEDUCTION
a) Heat a spatula end-full of **W** strongly in a dry test tube.		
b) To a spatula end-full **W** add 2-3 drops of		

conc.Sulphuric acid. Heat the mixture.		
c) Dissolve two spatula end-full of **W** in about $5cm^3$ of water and divide the solution into six parts.		
d) (i) To the first part of the solution, add dilute sodium hydroxide drop wise until in excess. Warm the mixture.		
ii. To the second part add dilute ammonia solution drop wise until in excess.		
iii. To the third portion add 2-3 drops of silver nitrate		
iv. To the fourth part add concentrated ammonia solution drop wise until in excess, then add hydrogen peroxide solution.		
v. To the fifth part, add 2-3 drops of lead (II) nitrate.		

vi. To the sixth carry out a test of your own to confirm the anion in **W.**		

Identify

i. Cations in **W.......................** and **.......................................**

(ii) The anions in **W. ....** and **.......................................**

Experiment 20

You are provided with substance **X** which contains two cations and an anion. Carryout the following tests on the solid **X** to identify the cations and anion in **X**. identify the gases evolved, record your observations and deductions in the table below.

Test	**Observation**	**Deduction**
(a) Dissolve a spatula end full of **X** in $4cm^3$ of distilled water. Divide the resulting solution into five portions.		
(i) To the first portion of the resulting solution add sodium hydroxide solution drop wise until in		

excess and warm the mixture.		
(ii) To the second portion of the resulting solution add ammonia solution drop wise until in excess.		
(iii) To the third portion of the resulting solution add potassium iodide solution.		
(iv) To the fourth portion of the resulting solution add a few drops of dilute nitric acid followed by lead (II) nitrate solution.		
(v) To the fifth portion of the resulting solution add silver nitrate solution.		

Identify the; cations………………………………………………………………………

Anion ……………………………………………………………………….

Write the equation for the reaction that took place in (v).

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Experiment 21

You are provided with substance **T** which contains two cations and an anion. Carryout the following tests on the solid **T** to identify the cations and anion in **T**. identify the gases evolved, record your observations and deductions in the table below.

Test	Observation	Deduction
(a) Heat a spatula end full of **T** strongly in a hard test tube until there is no further change.		
(b) Dissolve 3 spatula endfuls of **T** in 3cm^3 of nitric acid.		
(c) To the resulting solution from (b) add aqueous ammonia solution dropwise untill excess. Filter the mixture and keep both the filtrate and the residue.		
b) To the filtrate add a few drops of dilute nitric until the		

solution is just acidic. Divide the acidic solution into 2 portions i. To the first portion add sodium hydroxide solution dropwise until excess.		
ii. To the second portion of the filtrate add lead (II) nitrate solution.		
c) Dissolve the residue in a minimum amount of dilute nitric acid and divide the solution into three portions.		
(i) To the first portion add sodium hydroxide solution drop wise until in excess.		
(ii) To the second portion add ammonium hydroxide solution		

drop wise until in excess.		
(iii) To the third portion, add a few drops of potassium iodide solution.		

Identify the;

cations, ...

Anion ………………………………………………………………………..

Chapter 3: Rates of Reactions

In a chemical reaction, reactants are changed to products so that with time the amount of the reactants will be decreasing while the one of the products will be increasing. The rate of reaction will give information on how fast a given reaction takes place showing how much of the reactants is being used up or how much of the products are forming at a given time.

So the rate of the reaction can be expressed as the speed at which any reactant decreases or the speed at which any product increases.

The rate of a reaction is defined as" **the amount of product formed or amount of reactant used up per unit time**." Or **the speed at which reactants are turned into product.** The time taken for the reaction to take place must be in seconds.

For reactions involving solutions, the rate of reaction is often expressed as the rate of change of concentration of a reactant or product per unit time.

In summary, all rates are a quantity achieved at a given time so that the units of rate are quantities per unit time.

$$\text{Rate} = \frac{\text{measure of change in a given substance}}{\text{Time taken for the change to occur}}$$

The speeds of reactions are very varied.eg

- Rusting is a 'slow' reaction; you hardly see any change looking at it!
- The weathering of rocks is an extremely very slow reaction.
- The fermentation of sugar to alcohol is quite slow but you can see the carbon dioxide bubbles forming in the 'froth' in a laboratory experiment or beer making in industry!

- A faster reaction example is magnesium reacting with hydrochloric acid to form magnesium chloride and hydrogen or the even faster reaction between sodium and water to form sodium hydroxide.
- Combustion reactions e.g. when a fuel burns in air or oxygen, is a very fast reaction.
- Explosive reactions would be described as 'very fast' e.g. the pop of a hydrogen-air mixture on applying a lit splint.

The importance of "Rates of Reaction knowledge":

- Time is money in industry, the faster the reaction can be done, and the more economic it is.
- **Health and Safety Issues:** Mixtures of flammable gases in air present an explosion hazard e.g. Methane gas in mines, petrol vapour etc. are all potentially dangerous situations

A reaction will continue until one of the reactants is used up. To measure the 'speed' or 'rate' of a reaction depends on what the reaction is, and what is formed to be measured as the reaction proceeds. Two examples are outlined below.

When a gas is formed from a solid reacting with a solution, it can be collected in a gas syringe. The initial gradient of the graph e.g. in cm^3/s (speed or rate) gives an accurate measure of how fast a gaseous product is being formed in metal/carbonate- acid reaction (forming H_2/CO_2 respectively).

The gas formed is measured every e.g. 30 seconds and the graph plot measure the initial gradient in cm^3/sec.

The most accurate measurements are made early on in the reaction when the gas volume versus time is almost linear. You can take a series of measurements and draw the graph to get the rate from the gradient (e.g. cm^3/s) or measure the time to make a fixed volume of gas.

If the reaction is allowed to go on, you can measure the final maximum volume of gas and the time at which the reaction stops, though this a very poor measure of rate, because the reaction just goes slower and slower as the reactant amounts/concentrations are decreasing - so don't use this as a method of measuring reaction speed.

The reciprocal of the reaction time, 1/time, can also be used as a measure of the speed of a reaction. The time can represent how long it takes to form a fixed amount of gas.

The time can be in minutes or seconds, as long as you stick to the same unit for a set of results. Examples of reactions involving gas formation

i) Metals reacting with dilute acids, eg

$$Mg_{(s)} + H_2SO_{4(aq)} \longrightarrow MgSO_{4(aq)} + H_{2(g)}$$

(ii) Carbonates dissolving in acids to evolve carbon dioxide gas,

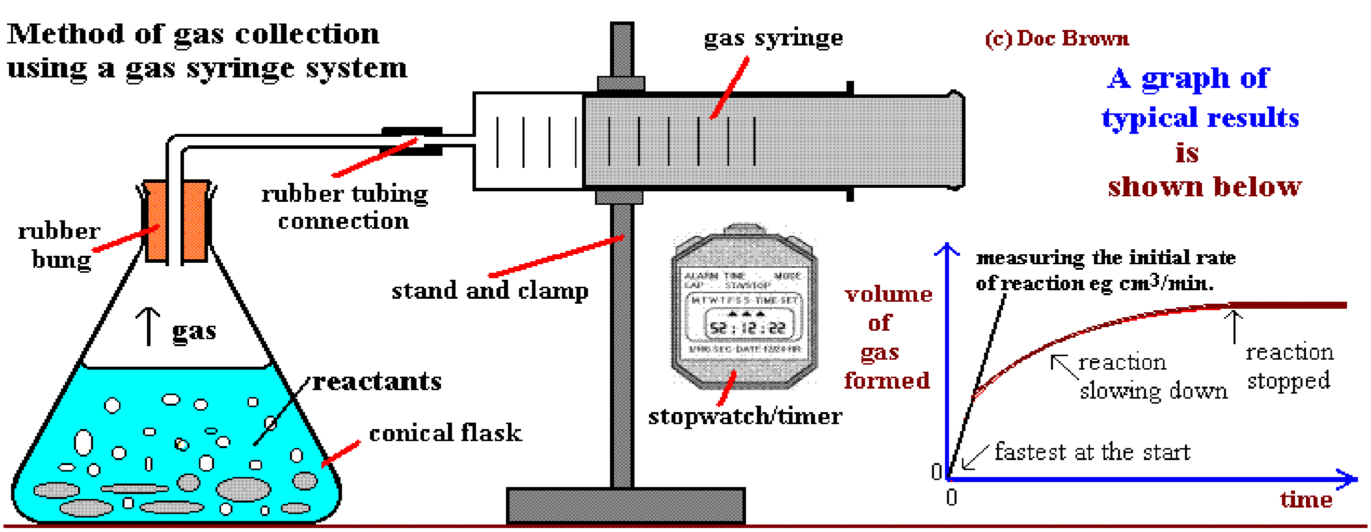
$$CaCO_{3(s)} + 2\ HCl_{(aq)} \longrightarrow CaCl_{2(aq)} + H_2O_{(l)} + CO_{2(g)}$$

(iii) The manganese (IV) oxide catalysed decomposition of hydrogen peroxide to form

oxygen.

$$2H_2O_{2(aq)} \longrightarrow 2H_2O_{(l)} + O_{2(g)}$$

These reactions can all be followed with the gas syringe method.



The shape of the graph is quite characteristic.

The reaction is fastest at the start when the reactants are at a maximum (steepest gradient in cm^3/s).

- The gradient becomes progressively less as reactants are used up and the reaction slows down.
- Finally the graph levels out when one of the reactants is used up and the reaction stops.
- The amount of product depends on the amount of reactants used.
- The initial rate of reaction is obtained by measuring the gradient at the start of the reaction. A tangent line is drawn through the first part of the graph, which is usually reasonably linear from the origin. This gives you an initial rate of reaction in cm^3 of gas/second,

How to calculate the reaction rate is explained below.

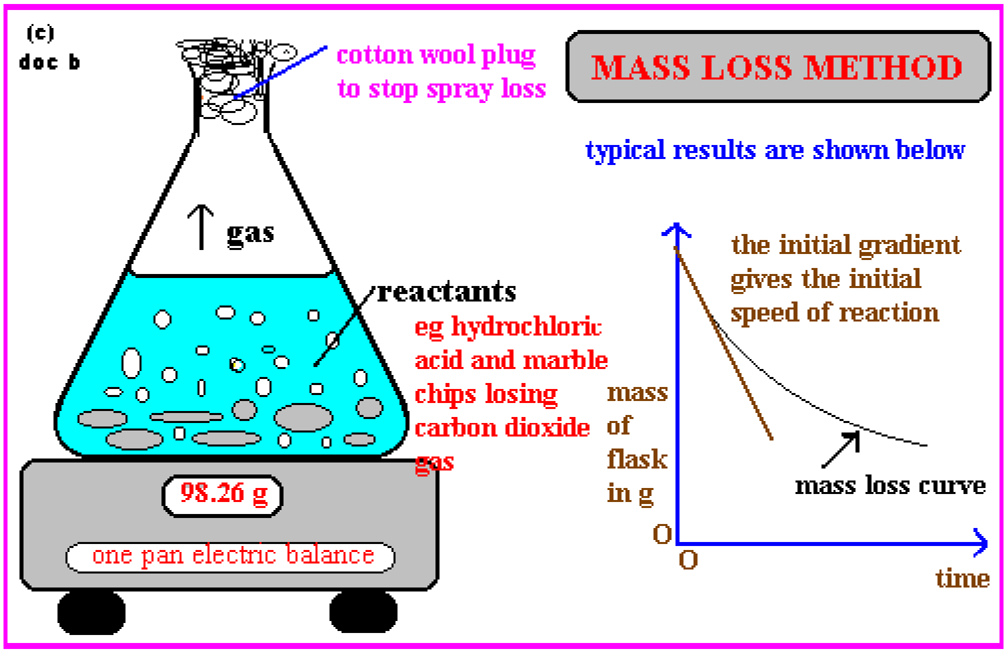
- From the graph of results you can measure the relative rate of reaction from
  (i) You can estimate from the graph the volume of gas formed after a particular time
  (ii) The initial gradient in cm^3/s

(iii) You can estimate the average rate between two points

- Keeping the temperature constant is really important if you are investigating speed of reaction/rate of reaction factors such as concentration of a soluble reactant or the particle size/surface area of a solid reactant.

The rate of a reaction that produces a gas can also be measured by following the mass loss as the gas is formed and escapes from the reaction flask.

- The method is good for reactions producing carbon dioxide or oxygen, but not very accurate for reactions giving hydrogen (too low a mass loss for accuracy).
- The reaction rate is expressed as the rate of loss in mass from the flask in e.g. g/s based on the initial gradient.



When sodium thiosulphate reacts with an acid, a yellow precipitate of sulphur is formed. To follow this reaction in your investigation you can measure how long it takes for a certain amount of sulphur to form.

This is done by observing the reaction down through a conical flask, viewing a black cross on white paper . The X is eventually obscured by the sulphur precipitate and the time noted.

$$Na_2S_2O_{3(aq)} + 2HCl_{(aq)} \longrightarrow 2NaCl_{(aq)} + SO_{2(aq)} + H_2O_{(l)} + S_{(s)}$$

By using the same flask and paper X you can obtain a relative measure of the speed of the reaction in forming the same amount of sulphur.

The speed or rate of reaction can expressed as amount of sulphur /time, so the rate is proportional to 1/time for a particular experiment. Since the absolute mass of sulphur formed is not known, the reciprocal of the time is taken as a measure of the relative rate of reaction.

The Factors affecting the Rate of Chemical Reaction.

The effect of Concentration

If the concentration of any reactant in a solution is increased, the rate of reaction is increased. This is because increasing the concentration, increases the probability of a collision between reactant particles because there are more of them in the same volume and so increases the chance of a fruitful collision forming products.
e.g. Increasing the concentration of acid molecules increases the frequency or chance at which they hit the surface of marble chips to dissolve them

The effect of Pressure

If one or more of the reactants is a gas then increasing pressure will effectively increase the concentration of the reactant molecules and speed up the reaction.
The higher pressure of a gas results in greater concentration and so faster reaction all because of the increased chance of a 'fruitful' collision.

The effect of Surface Area - particle size of a solid reactant

- If a solid reactant or a solid catalyst is broken down into smaller pieces the rate of reaction increases.
- This is because small pieces of marble have greater surface area for attack by the acid. The speed increase happens because **smaller pieces of the same mass of solid have a greater surface area** compared to larger pieces of the solid

The effect of Temperature

- When gases or liquids are heated the particles gain kinetic energy and move faster
- The increased speed increases the chance (frequency) of collision between reactant molecules and the rate increases.
- This means that the increased chance of 'fruitful' higher energy collision greatly increases the speed of the reaction, depending on the fraction of molecules with enough energy to react

The effect of a Catalyst

- The word catalyst means an added substance, in contact with the reactants, that changes the rate of a reaction without itself being chemically changed in the end.
- Catalysts increase the rate of a reaction by helping break chemical bonds in reactant molecules and provide a 'different pathway' for the reaction.
- Therefore at the same temperature, more reactant molecules have enough kinetic energy to react compared to the uncatalysed situation. The catalyst does not increase the energy of the reactant molecules.
- True catalyst does not change chemically or get used up in the end, can be reused/regenerated with more reactants. Examples
  - ✓ Black manganese (IV) oxide catalyses the decomposition of hydrogen peroxide.

$$\mathbf{2H_2O_{2(aq)} \longrightarrow 2H_2O_{(l)} + O_{2(g)}}$$

  - ✓ Nickel catalyses the hydrogenation of unsaturated fats to margarine
  - ✓ Iron catalyses the combination of unreactive nitrogen and hydrogen to form ammonia.
  - ✓ Enzymes in yeast convert sugar into alcohol

- Enzymes are biochemical catalysts. They have the advantage of bringing about reactions at normal temperatures and pressures which would otherwise need more expensive and energy-demanding equipment.

3 The Effect of Light

Light energy can initiate or catalyse particular chemical reactions. . The greater the intensity of light (visible or ultra-violet) the more reactant molecules are likely to gain the energy react, so the reaction speed increases.

Examples:

- Silver salts are converted to silver in the chemistry of photographic exposure of the film.
  - ✓ Silver chloride, silver bromide and silver iodide are all sensitive to light ('photosensitive'), and all three are used in the production of various types of photographic film to detect visible light.

- ✓ Each silver halide salt has a different sensitivity to light.
- ✓ When radiation hits the film the silver ions in the salt are reduced by electron gain to silver

  Ag^{+} (aq) + e^{-} $\longrightarrow$ Ag(s)

(X = halogen atom, Cl, Br or I) and the halide ion is oxidised to the halogen molecule by electron loss

- $2Cl^{-}$(aq) $\longrightarrow$ Cl_2(g)+ $2e^{-}$

so overall the change via light energy is

$2AgCl$ (aq) $\longrightarrow$ 2Ag(s) + $Cl_{2(g)}$

- AgI is the least sensitive and used in X-ray radiography, AgCl is the most sensitive and used in 'fast' film for cameras.
- Photosynthesis in green plants requires the input of sunlight energy and the green chlorophyll molecules absorb the energy light.
  - $6H_2O_{(l)} + 6CO_{2(g)} \longrightarrow C_6H_{12}O_{6(aq)} + 6O_{2(g)}$

Experiment one

1. You are provided with the following;

   BA_3 which is a solution containing of sodium thiosulphate solution.

   BA_4 which is a 2.0 M solution of hydrochloric acid. You are required to investigate the effect of concentration on the rate of reaction

   Procedure.

(i) Measure 60 cm^3 of sodium thiosulphate in a clean beaker

(ii) Place the beaker on a piece of paper marked with a cross. The cross should be seen through the solution before beginning the experiment

(iii) Add 5 cm^3 of hydrochloric acid using a measuring cylinder, swirl the mixture and start the stop clock at the same time.

(iv) Record the time taken for the cross to disappear in a table.

(v) Repeat steps (i) – (ii) using volumes given in the table.

(vi) Compute the values of 1/t.

Results

Experiment	Volume of thiosulphate (cm^3)	Volume of water added(cm^3)	Volume of hydrochloric acid added (cm^3)	Time taken for cross to disappear(s)	Rate, 1/t (s^{-1})
1	60	0	5		
2	50	10	5		
3	40	20	5		
4	30	30	5		
5	20	40	5		

a) Plot a graph of 1/t against volume of sodium thiosulphate.

b) Write the equation for the reaction between hydrochloric acid and Sodium thiosulphate.

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c) What is the effect of changing the concentration on this rate of reaction?

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d) Calculate the slope of your graph and state its units.

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Experiment two: to investigate the effect of surface area on the rate of reaction

You are provided with the following;

A 2M hydrochloric acid, Marble chips,

You are required to determine the effect of surface area on the rate of the reaction

Procedure

Method

a) Weigh 10.0 g of marble chips on a balance

b) Using a measuring cylinder measure $100cm^3$ of the acid and pour it into the conical flask.

c) Transfer the marble chips onto the acid in the flask, quickly place the flask on the balance and start the clock at the same time. Ensure that you close the mouth of the flask with a plug of cotton wool.

d) Note the changes in mass as carbon dioxide is given off at intervals of 30 seconds

e) Record your results in the table below.

f) Crush an equal mass of marble chips to powder using a mortar and pestle and repeat steps (a)- (e)

Table of results

Time (s)	30	60	90	120	150	180	210
Marble chips.(20g)							
Marble powder.(20g)							

Questions

(a) Plot a graph of mass of flask and contents against time on the same graph.

(b) From your graph, determine the mass of the flask containing marble chips and powder after 80 seconds.

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(c) Explain how surface affects the rate of reaction

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Questions

1. (a) What is meant by the term *Rate of a chemical reaction*?

   ii) State and explain any three factors that affect the rate of a chemical reaction.

   b) A specific volume of 0.2M sulphuric acid was added to excess magnesium powder, resulting into evolution of $120cm^3$ of a gas at room temperature and pressure.

   (i) Draw a labelled diagram of the apparatus that can be used to collect the gas and the gas volume used to determine the rate of the reaction.

   (ii) Write an equation for the reaction.

   d) Calculate the

   (i) Mass of the magnesium powder used in the reaction.

   (ii) Volume of 0.2M sulphuric acid added in the reaction.

   e) (i) Draw a graph of volume of hydrogen against time when excess Magnesium powder to the same volume of 0.2M sulphuric acid and label the graph P.

   (ii) On the same axes draw a graph of volume of hydrogen against time when excess magnesium ribbon is added to the same volume of 0.2M sulphuric acid and label the graph Q

(ii) Explain the difference between the two graphs.

2. The results below are obtained during the reaction between excess powder calciumcarbonate and dilute hydrochloric acid.

volume of CO_2/ cm^3	0	20	35	47	56	64	69	73	77	79	80	80
Time /seconds	0	10	20	30	40	50	60	70	80	90	100	110

(i) Plot a graph of volume of carbon dioxide evolved against time

(ii) Explain the shape of the graph.

(iii) Determine the rate of evolution of CO_2 at 20 and 50 seconds and explain the difference in the two rates.

(iv) How long did it take for $40cm^3$ of carbon dioxide to be formed?

(v) What volume of gas was formed after 15 seconds?

Chapter 4: ENERGY CHANGES IN CHEMICAL REACTIONS.

Fuels are Substances which give out heat and Light when they burn. This energy is used in many ways by man- These include; cooking, keeping us warm, drives motor engines etc. Most of the energy comes from fossil fuels like coal, oil and natural gas. Wood and charcoal are the most common Sources of energy used in Uganda.

These fuels except Carbon contain hydrogen and Carbon; they give out water and carbon dioxide plus energy when they burn. A good fuel should be:

- cheap,
- burn easily,
- be easily and safely stored,
- burn without too much Smoke, solid residues or pollutant gases,
- be easily transported with little or no risk

Enthalpy changes.

Enthalpy is the energy /heat content of substance which is stored in its bond. Its Symbol is H. This energy is difficult to measure. What is easily measured is the change in enthalpy, denoted by ΔH

When a reaction occurs some of the old bonds are broken in the reactants and new bonds are formed in the products

Bond breaking absorbs energy while bond formation gives out energy to the surroundings. It is the energy difference in bond breakage and bond formation that is measured as enthalpy change. This is measured by a fall or rise in temperature on the thermometer.

Endothermic and Exothermic reactions.

Exothermic reaction is the reaction is one in which heat is given to surroundings. The temperature rise is the measure of this energy. The energy of products is less than that of reactants.

Enthalpy change of this reaction is given by $\Delta H = H_p - H_r$, where H_p is enthalpy of products and H_r is the enthalpy of reactants. Thus ΔH is negative.

An endothermic reaction is reaction in which heat is absorbed from the surroundings. The temperature drop is the measure of this energy. The energy of the products is more than that of reactants, therefore ΔH is positive.

The units of heat of reaction are kJ mol^{-1} for a specified reactant or product.

Types of enthalpies

Enthalpy of combustion

Heat of Combustion of a substance is the heat liberated when 1 mole of the substance undergoes complete combustion with oxygen at constant pressure. Combustion is always exothermic, △H is negative.

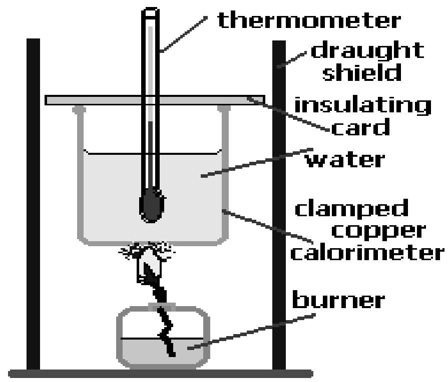
$$CH_{4(g)} + 2O_{2(g)} \longrightarrow CO_{2(g)} + 2H_2O_{(l)} \qquad \triangle H = -890 \text{ kJ mol}^{-1}$$

1 mole of methane gas, $CH_{4(g)}$, undergoes complete combustion in excess oxygen gas releasing 890 kJ of heat. Heat of Combustion can be measured experimentally.

Measuring Heat of Combustion

- A known quantity of water is placed in a flask or beaker.
- A thermometer is positioned with bulb near the middle of the volume of water.
- A known quantity of fuel, such as an alcohol (ethanol), is placed in the spirit burner.

- The initial temperature of the water is measured and recorded (T_i). The wick on the spirit burner is lit, burning the fuel, and heating the water.
- When the temperature has risen an appreciable amount, the spirit burner is extinguished and the final temperature recorded (T_f).
- The final quantity of fuel is measured and recorded.



Typical Results for an experiment where ethanol is used to heat 200g of water:

initial water temperature (T_i) = 20^oC	intial mass burner + ethanol = 37.25 g
final water temperature (T_f)= 75^oC	final mass burner + ethanol = 35.50 g
change in temperature = T_f - T_i = 55^oC	mass ethanol used = 1.75g

Calculation of Heat of Combustion of ethanol:

✓ Calculate moles (n) of fuel used

Molecular mass of ethanol = 46.1g
mass ethanol used = 1.75g

Number of moles = mass, M = 1.75 ÷ 46.1 = 0.0380 mol

✓ Calculate energy required to change temperature of water.

Energy = specific heat capacity of water x mass of water x change in temperature

Energy = 4.184 x 200 x 55

= 46024 J

= 46.024 kJ

✓ Calculate the heat of combustion of ethanol

Assume all the heat produced from burning ethanol has gone into heating the water, i.e. no heat is lost.

0.0380 mole ethanol produced 46.024 kJ of heat.

Therefore 1 mole of ethanol would produce $\frac{46.024\text{kj}}{0.0380 \text{ mol}}$

= 1211 kJ/mol

The heat of combustion of ethanol is 1211 kJ/mol

1) 100 cm^3 of water (100g) was measured into the calorimeter. The spirit burner contained the fuel ethanol C_2H_5OH ('alcohol') and weighed 18.62g at the start. After burning it weighed 17.14g and the temperature of the water rose from 18 to 89^{o}C. (C = 12, H = 1, O = 16)

Solution

The temperature rise = 89 - 18 = 71^{o}C (exothermic, heat energy given out).

Mass of fuel burned = 18.62-17.14 = 1.48g.

Heat absorbed by the water = mass of water x SHC_{water} x temperature

= 100 x 4.2 x 71 = 29820 J (for 1.48g)

Heat energy released per g = energy supplied in J / mass of fuel burned in g

Heat energy released on burning = 29820 / 1.48 = 20149 J/g of C_2H_5OH

This energy change can be also expressed on a molar basis.

Molar mass (C_2H_5OH) = (2 x 12) + (1 x 5) + 16 + 16 = 46, so 1 mole = 46g

Heat released by 1 mole of C_2H_5OH = 46 x 20149 = 926854 J/mole or 927 kJ/mol (3 sf)

Enthalpy of combustion of ethanol = -927 kJmol^{-1}

This means 926.9 kJ of heat energy is released on burning 46g of ethanol

The experimentally determined value for the heat of combustion of ethanol is usually less than the accepted value of 1368 kJ/mol because some heat is always lost to the atmosphere and in heating the vessel.

Experiment to determine the heat of combustion of ethanol.

Material required.

- Tin can, Thermometer, Spirit burner, Card board, beaker, Tripod stand, Ethanol, Water, Measuring cylinder, Pipe clay triangle**.**

Procedure.

1. Fill the burner half full with ethanol,
2. Weigh the spirit burner with its cover and record its mass.
3. Measure 100cm^3 of water and pour it into the metal can.
4. Measure the temperature of water.
5. Set up the experiment as shown in the set up below
6. Surround the set up with the cardboard, with one side cut out to allow access.

7. Heat the water until the water temperature rises by $20^{O}C$, then extinguish the flame on the burner and cover the wick.
8. Record the final temperature of the water. And reweigh the spirit burner with its cover.

Results:

Initial mass of the burner and ethanol = ………………………………………………g

Final mass of the burner and ethanol = …………………………………………….. g

Mass of ethanol burnt = …………………………………………….......g

Final temperature of water = ……………………………………………….. ^{O}C

Initial temperature of water =……………………………………………… ^{O}C

Temperature rise = ……………………………………………………… ^{O}C

Questions.

1) Calculate

    a) The number of moles of ethanol used. (C=12, H= 1, O= 16)

b) The energy given out/ gained by water.

(Specific heat capacity of water $=4.2Jg^{-1}K^{-1}$)

c) The energy given out by one mole of ethanol.

2) Write the equation for the combustion of ethanol showing its energy notation.

3) The standard enthalpy of combustion of ethanol is $-1371kJ\ mol^{-1}$.Why is this value different your experimental value?

4) (i) Why was the metal can used instead of a glass beaker?

(ii) What was the purpose of the cardboard?

Enthalpy of neutralization.

The enthalpy of neutralization is the heat change that occurs when an acid reacts with a base to form one mole of water and a salt.

$$NaOH_{(aq)} + HCl_{(aq)} \longrightarrow NaCl_{(aq)} + H_2O_{(l)},\ \Delta H = -57.1\ kJ\ mol^{-1}$$

$$½Ba(OH)_{2(aq)} + HNO_{3(aq)} \longrightarrow Ba(NO_3)_{2(aq)} + H_2O_{(l)}, \Delta H = 58.2\ -kJ\ /mol$$

The heat of neutralization for strong acids and strong base is usually 57.1 kJ/mo1. This is because the strong acids and bases completely dissociate into ions when dissolved in water. If the acid or base is weak, the enthalpy will be less than 57.1 kJ mol^{-1}. This is because some energy is absorbed to dissociate them into ions. An example of weak acid is ethanoic acid, ammonia is a weak base.

Measuring Heat of Reaction Experimentally

- A known quantity of reactant is placed in a well insulated vessel (eg, a plastic cup)
- The initial temperature of this reactant is recorded, T_i.
- A known quantity of the second reactant is added, the vessel is sealed with a lid and the reaction mixture stirred.
- The final temperature of the reaction mixture is recorded, T_f.
- The heat released or absorbed, in joules, for the reaction is calculated:

  q = mass x specific heat capacity x change in temperature

- The enthalpy change in kJ per mole of a given reactant for the reaction is calculated: H = q/1000 ÷ moles

Common assumptions for reaction mixtures made up of aqueous solutions:

- ✓ density of aqueous solution assumed to be the same as for water, $1 g/cm^3$

- ✓ volumes of reactants is added e.g., 100cm^3 of reactant a + 200cm^3 of reactant b = 300 cm^3 of reaction mixture
- ✓ specific heat capacity of the reaction mixture assumed to be the same as that of water, i.e. specific heat capacity = 4.2 $JK^{-1}g^{-1}$
- ✓ Heat is not lost to, or absorbed by, the surroundings.

Experiment to determine the enthalpy of neutralization of hydrochloric acid by sodium hydroxide

Apparatus and chemicals.

- 2 beakers, Plastic beaker/ cup, Cotton wool, old newspaper, Thermomete, 1M hydrochloric acid, 1M sodium hydrochloric.

Proceure.

1. Measure 50cm^3 of hydrochloric acid and pour it into a beaker.
2. Measure 50cm^3 of sodium hydroxide solution into another beaker.
3. Take and record the temperature of sodium hydroxide and hydrochloric acid.
4. Pour the acid and the alkali into the insulated plastic beaker. Stir mixture and record the final temperature.

Results.

Temperature of hydrochloric acid T_1 =... ^oC

Temperature of sodium hydroxide solution, T_2 = ... ^oC

Average temperature, $\frac{T_1 + T_2}{2}$ = ... ^oC

Temperature of the mixture, T_3 = ... ^oC

Questions:

a) Calculate the temperature change, ΔT.

b) Determine the energy given out in kJ.

c) Calculate the number of moles of water formed.

d) Write the ionic equation for the reaction that took place.

e) Calculate the molar heat of neutralization for this reaction.

3. An experiment to determine the molar heat of neutralization by graphical method.

You are provided with the following:

W, which a solution of a potassium hydroxide.

X, which is a solution of an acid.

You are required to determine the heat of neutralization between potassium hydroxide and the acid.

Pocedure.

a) Using a measuring cylinder, measure 150cm^3 of W and transfer into 250cm3 beaker. Add 50cm^3 of distilled water, mix and label it BA1
b) Transfer 100cm3 of X into another 250 beaker using a measuring cylinder. Add 100 cm^3 of didtilled water, mixand label it BA2.
c) Measure and record the intial temperature of BA1
d) Run 25.0 cm^3 of BA1 from the burette into a dry plastic beaker.
e) Using a measuring cylinder, transfer at once 10cm^3 of BA2 into the plastic beaker containing BA1.
f) Repeat the procedure (d) and (e) using 20, 30,40 and 50cm^3 of BA2.
g) Record your results in the table below

Initial temperature of BA1……………………………..

Volume of BA2 used (cm^3)	10	20	30	40	50
Highest temperature attained by mixture (^{O}C)					

Questions

(a) Plot the graph of highest temperature attained by the mixture against volume of BA2 used.

(b) From the graph, determine:

(i) The volume of BA2 required to neutralize $25.0cm^3$ of BA1

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(ii) The maximum temperature change for the reaction.

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(c) Calculate the molarity of BA2(mole ratio of BA1:BA2 is 1:1, Molarity of BA1 is 1.5M)

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(i) Determine the heat evolved during the reaction
(Specific heat capacity of the mixture = $4.2Jg^{-1}K^{-1}$; density of mixture = $1gcm^{-3}$)

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(ii) Molar heat of reaction between the base and the base.

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Example

1. When 25 cm^3 of 0.700 M NaOH was mixed in the same calorimeter with 25 cm^3 of 0.700 M HCl both initially at 21 degrees C, the temperature increased to 25.1 degrees C.

$NaOH_{(aq)}$ + $HCl_{(aq)}$ $\longrightarrow$ **NaCl $_{(aq)}$ + $H_2O_{(l)}$** + heat

Total volume of the mixture = (25+25) cm^3

Mass of mixture =50 x1 =50 g

Heat gained by solution= m x shc x ΔT = 50 x 4.184 x (25.1-21.0) =861J

Mole of the acid or base that reacted = $\frac{0.700 X 25}{1000}$ = 0.0175

0.0175 moles of acid give out 861 J on complete neutralization,

Therefore 1 mole gives out $\frac{861}{0.175}$ J = 49200J =49.2kJ

Heat of neutralization is -49.2kJ per mole

2 In a coffee cup calorimeter, 100.0 cm^3 of 1.00 M NaOH and 100.0 cm^3 of 1.00 M HCl are mixed. Both solutions were originally at 24.6 °C. After the reaction, the final temperature is 31.3 °C. The density of 0.500 M NaCl is 1.02 g/ cm^3, the specific heat capacity (s) of 0.500 M NaCl(aq) is 3.87 J/g°C

$NaOH_{(aq)}$ + $HCl_{(aq)}$ $\longrightarrow$ **NaCl $_{(aq)}$ + $H_2O_{(l)}$** + heat

Heat lost by the reaction = heat gained by the NaCl solution

The volume of NaCl (mixture) = 100+100 = 200 cm^3

The mass of the NaC l= (200 X1.02) g

Mass = 204g

ΔT for the NaCl = 31.3 - 24.6, ΔT =6.7^{o}C,

Heat gained = shc x m xΔT

= 204 x3.87 x 6.7

=5289.516 J

The heat gained by the solution is equal but opposite in sign of the heat lost by the reaction.

The heat lost =5289.516 J

The number of moles of either NaOH or HCl that reacted = $\frac{100X1}{1000}$ =0.1

0.1 moles give out 5289.516J of energy, hence one mole gives out $\frac{5289.516}{0.1}$

=52895.16J

=52.89kJ

Heat of neutralization= - 52.89 kJ/mole

The heat of reaction for a solute dissolving in a solvent is known as the **heat of solution.**

Experiment to determine the heat of solution of ammonium nitrate.

PROCEDURE

1. Measure 100cm^3 of distilled water using a measuring cylinder and pour it into a Plastic bottle.
2. Record the initial temperature of the water.
3. Measure 8g of ammonium nitrate and to the water cork the bottle and shake it gently to make a solution
4. Note the highest or lowest temperature of the solution.

Results:

Initial temperature of water = ... ^OC

Final temperature of solution ....^OC

Temperature change = ... ^OC

a) Determine the mass of ammonium nitrate formed.

(Volume of water =100cm^3, density of solution =1.0g/cm^3)

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b) Calculate the number of moles of ammonium nitrate that dissolved.

(N = 14, H = 1, O =16)

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c) Enthalpy of solution of ammonium nitrate in kJmol^{-1}

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Experiment to determine enthalpy of displacement

You are provided with the following;

- BA5; which is a solution containing $45gl^{-1}$ of hydrated copper (II) sulphate, $CuSO_4.5H_2O$
- Solid T

You are required to determine the molar enthalpy of displacement of copper from copper (II) sulphate using solid T

Procedure:

(a) Using a measuring cylinder, measure $40cm^3$ of solution BA5 into a $100cm^3$ plastic beaker. Note the initial temperature of the solution and record it in the table in the first column under time = 0

(b) Add all solid T at once to the solution in the beaker and start the stop clock immediately. Stir the mixture continuously with the thermometer as you record the temperature of the solution after every half a minute for seven minutes in the table of results below

Time/min	0	½	1	1½	2	2½	3	3½	4	4½	5	5½	6	6½	7
Temperature ^oC.															

Questions:

(a) Plot a graph of temperature (vertical axis) against time (horizontal axis)

(b) Use the graph to determine the highest temperature change ΔT

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(c) Calculate the

(i) Amount of heat given out during the reaction (specific heat capacity of solution = 4.2 $Jmol^{-1} k^{-1}$, density of solution = $1gcm^{-3}$)

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(ii) Molar enthalpy of displacement (Cu = 64 , S = 32, O = 16, H = 1)

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Questions

1. When 25 cm^3 of 2M sodium hydroxide was mixed with 25 cm^3 of 2M hydrochloric acid n a plastic cu, a temperature rise of 12.5^{0}C was recorded.

   (a) State whether the reaction is exothermic or endothermic.
   (b) Write an ionic equation for the reaction.
   (c) Calculate the heat of neutralization.
   (Density of solution = 1 g cm^{-3}) (S.H.C. of a dilute solution = 4.2 $Jg^{-1}\,^oC^{-1}$)

2. 20 cm^3 of 4MH_2SO_4 at 22^{0}C was added to 20 cm^3 of 2M sodium hydroxide solution at 23^{0}C. The final temperature was 35.5^{0}C. (S.H.C. of a dilute solution = 4.2 $Jg^{-1}\,^0C^{-1}$)

   (a) Write ionic equation for the reaction.

(b) Determine the enthalpy of neutralization.

3. The table below shows a change in temperature as hydrochloric acid is added to 25 cm^3 of 2M sodium hydroxide solution.

Volume of NaOH (cm^3)	25	25	25	25	25	25
Volume of HCl (Cm^3)	0	10	20	30	40	50
Temperature ^{0}C.	25	33	39	42	39	36

   (a) Plot a graph for temperature against volume of hydrochloric acid.
   (b) Determine the molarity of hydrochloric acid.

   (c) Determine the enthalpy of neutralization for the reaction.

   (Density of solution = 1 g cm^{-3}, S.H.C. = 4.2 J $g^{-1}\,^0C^{-1}$)

   (d) The experiment was repeated using ammonia solution instead of sodium hydroxide solution. State whether the maximum temperature would be equal to, greater than or less than that indicated on the graph. Explain your answer.

4. When 10g of ammonium chloride was dissolved in 100g of water, the temperature fell by 6^{0}C. Determine the heat of solution of ammonium chloride.

   (S.H.C. of water = 4.2 Jg^{-1} K^{-1}, N = 14, H = 1, Cl = 35.5)

5. Describe an experiment by which the heat of neutralization between 20 cm^3 of 2M hydrochloric acid and 20 cm^3 of 2M sodium hydroxide solution can be determined.